



Deliverables:

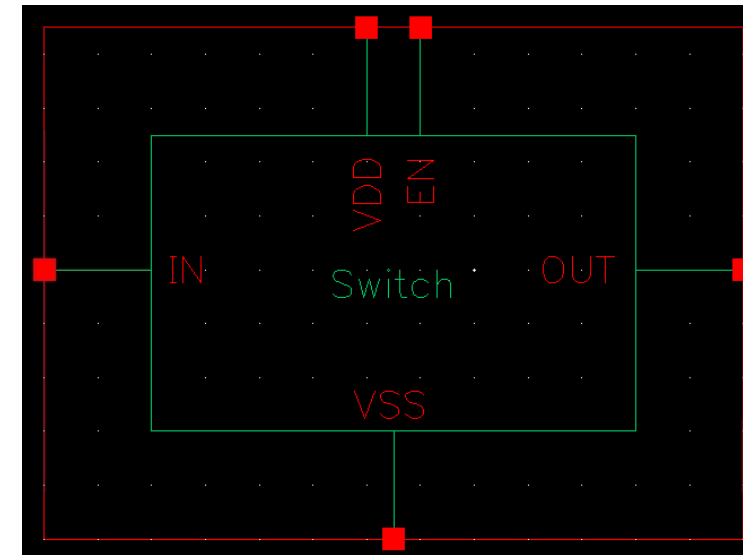
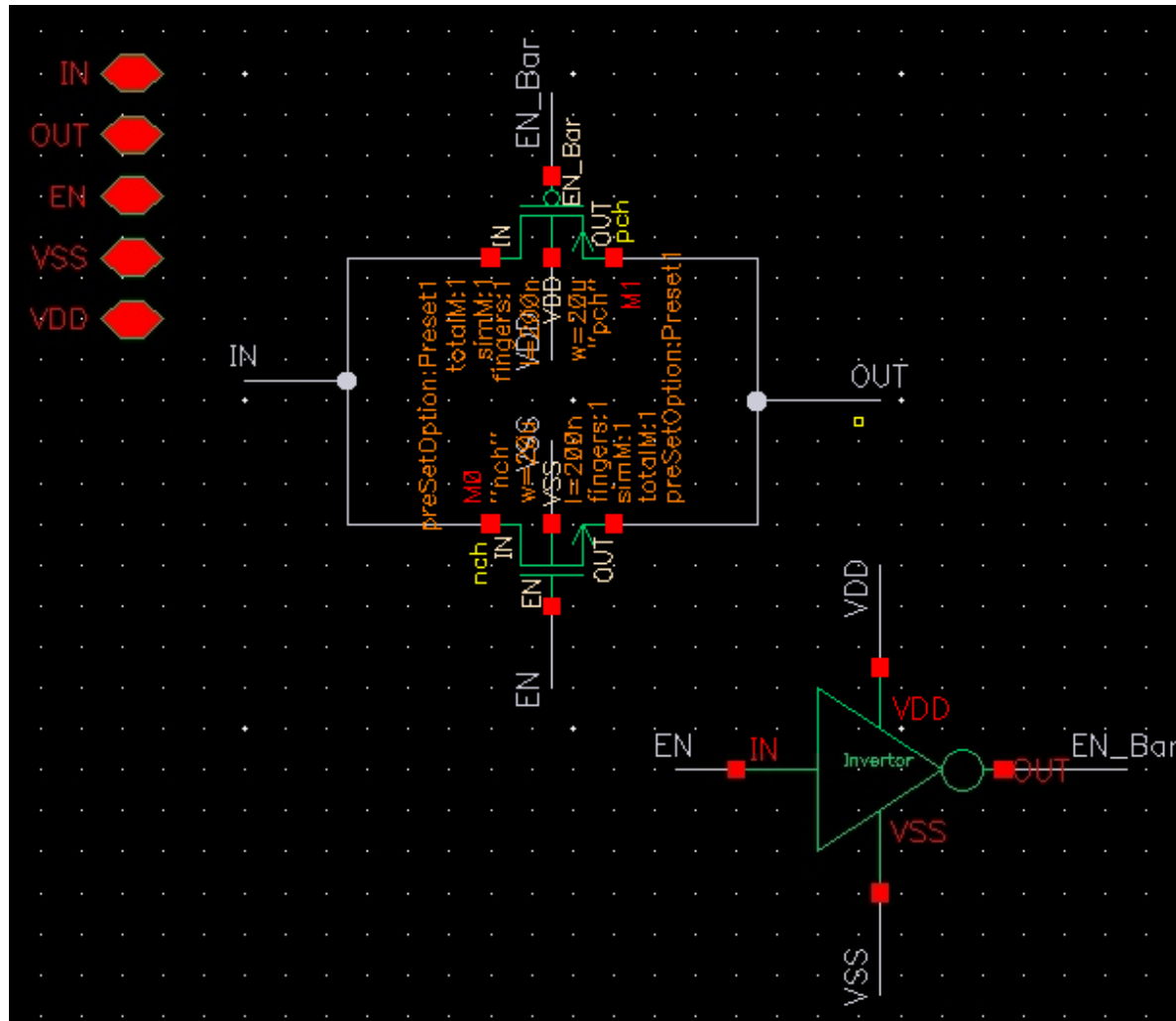
Code Checker
Transmission Gate
Differential Amplifier
Layout

Abstract

The purpose of the project was to design an amplification module to ensure quality amplification, ensure security and consistent activation among other amplification modules, and learn best practices during the layout process for fabrication.

A single-ended output differential amplifier sub-module was created to produce >26 dB gain at 1Mhz while maintaining the minimal power consumption (<500 μ W) and maximal peak-to-peak swing (>300 mV). Secondly, a transmission gate, to facilitate activation of module, was created with respectable on and off resistances ($R_{on} < 100$ and $R_{off} > 10M$). Lastly, the code checker (utilized NAND logic) was created to drive activation of the module within a timely matter, with a constrained propagation delay (≤ 500 ps overall and ≤ 15 ps per cell) and unit rise/fall time (<15 ps). The entire ensemble needed to maintain the amplification of >26 dB gain and meet quiescent current requirements during activation and de-activation.

During the layout process, layer selection was carefully selected and managed to provide signal integrity and keep manufacturability, for example higher layers were used to ensure connectivity between long distances and lower layers were used for local connections, and oscillation between vertical and horizontal directions ensured minimal parasitic capacitance.





Analyses

dc

t 100m 1.2 100 Linear Number of Steps Start-Stop

Click to add analysis

Design Variables

A

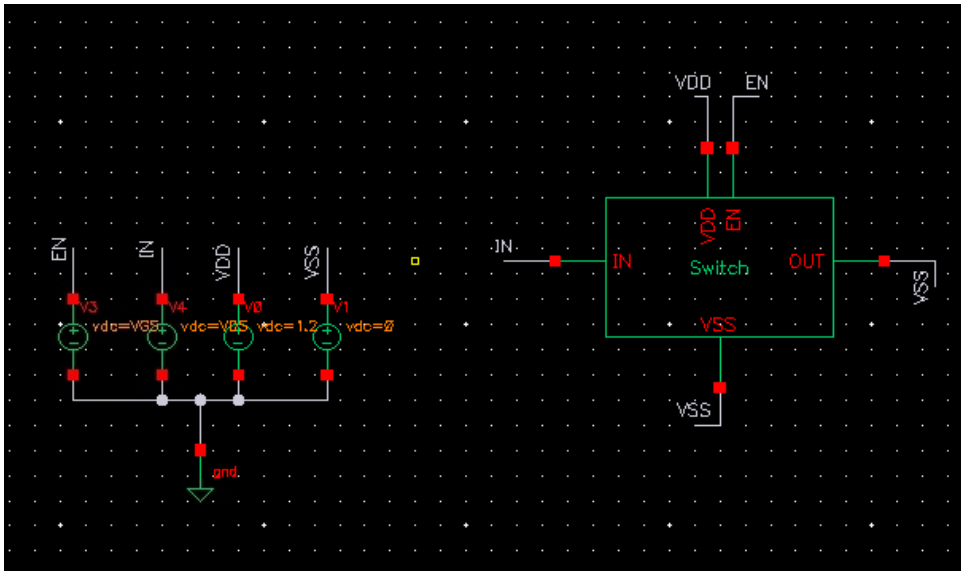
100n

VDS

1.2

VGS

0



Transmission Gate Off resistance, simulation setup, and result

Test	Name	Type	Details	EvalType	Plot	Save	Spec
Filter	Filter	Filter	Filter	Filter			Filter
Proj_208_Transmission_Gate_OFF_TB_1	ids	expr	IDC("I4/MINUS")	point	☒	☐	
Proj_208_Transmission_Gate_OFF_TB_1	vds	expr	VDC("IN")	point	☒	☐	
Proj_208_Transmission_Gate_OFF_TB_1	roff	expr	(vds / ids)	point	☒	☐	> 10M

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
Proj_208_Transmission...	ids	137.3p			
Proj_208_Transmission...	vds	1.2			
Proj_208_Transmission...	roff	8.74G	> 10M		pass

Explanation:

The above table highlights the current and the voltage going through the transmission gate when simulated to be off, which can be divided to receive the off resistance. The transistor dimensions of $I = 200\text{n M}$ and $W = 20\text{u M}$ leads to the off resistance to be 8.74G , which is beyond the requirement of greater than 10M ohms



Transmission Gate On resistance, simulation setup, and result

Analyses

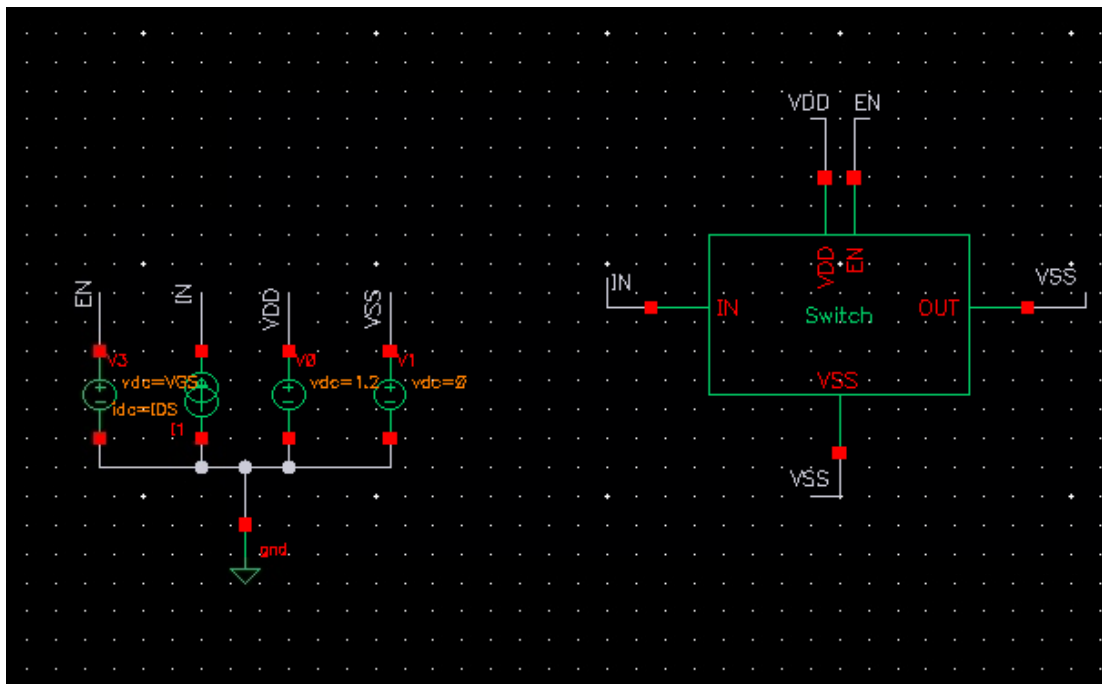
☒ dc t 1u 1m 30 Logarithmic Points Per Decade Start-Stop

Click to add analysis

Design Variables

A	100n
IDS	1m
VGS	1.2

Name	Type	Details	Value	Plot	Save	Spec
ids2	expr	IDC("/I1/PLUS")		☐	☐	
vds2	expr	VDC("/IN")		☐	☐	
ron	expr	average((vds / ids))	59.94	☒	☐	< 100
vds	expr	v("/IN" ?result "dc")		☒	☐	
ids	expr	i("/I1/PLUS" ?result "dc")		☒	☐	



Explanation:

The above table highlights the current and the voltage going through the transmission gate when simulated to be on, which can be divided to receive the on resistance. The transistor dimensions of $I = 200\text{nM}$ and $W = 20\text{uM}$ leads to the on resistance to be 59.94 ohms, which is beyond the requirement of less than 100 ohms

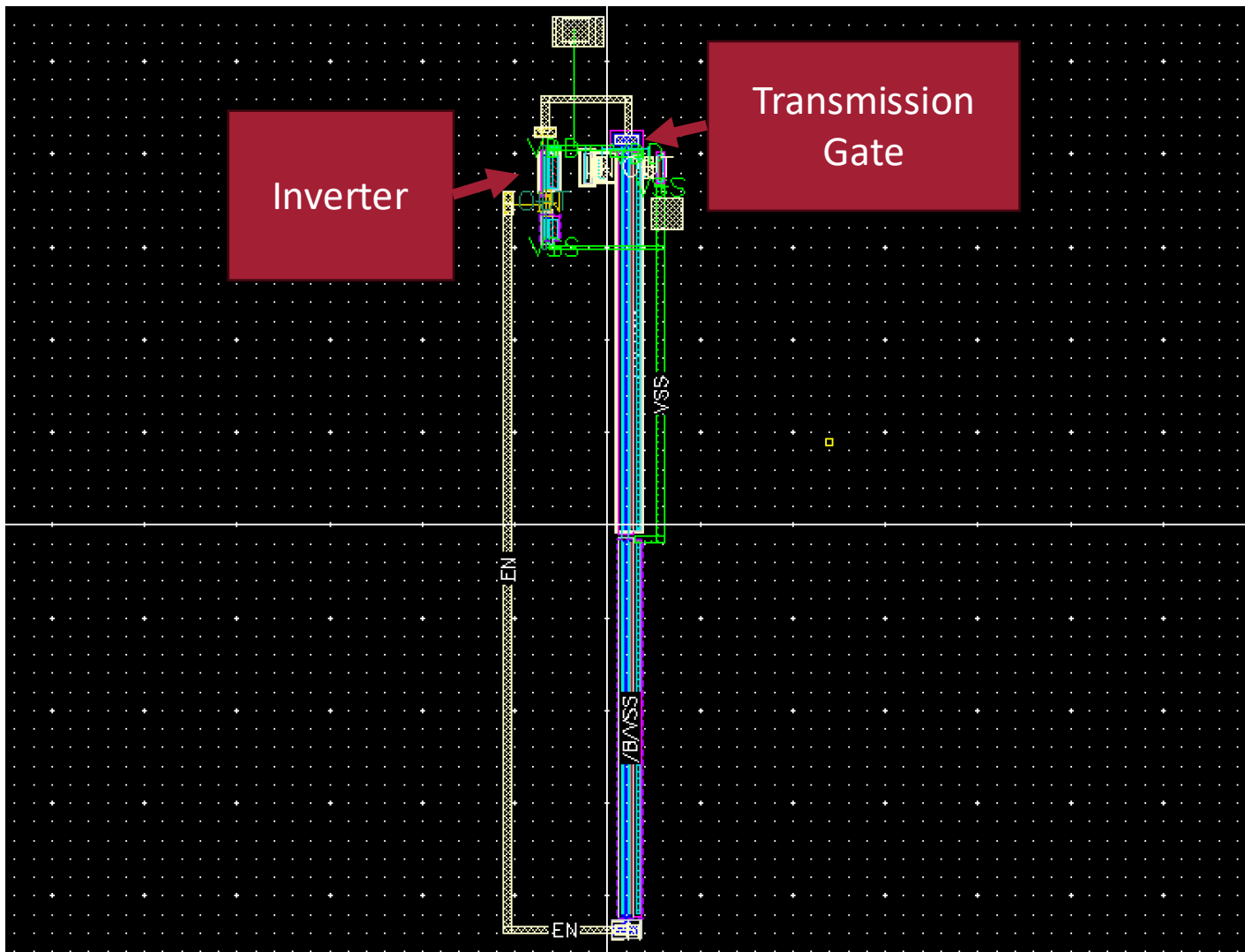


Image of Layout & DRC/LVS Clean

Calibre - RVE v2025.3_28.17 : Transmission_Gate.drc.results

File View Highlight Tools Window Setup Help

Filter: Show All Transmission_Gate, 1 Result (in 1 of 5 Checks)

Check / Cell	Results
Check DENSITY_PRINT_FILES	0
Check DRM.R.1	1

Comparison Results

Layout Cell / Type	Source Cell	Nets
Transmission_Gate	Transmission_Gate	6L, 6S

Cell Transmission_Gate Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

```
#####  
#      #  
# CORRECT #  
#      #  
#####
```

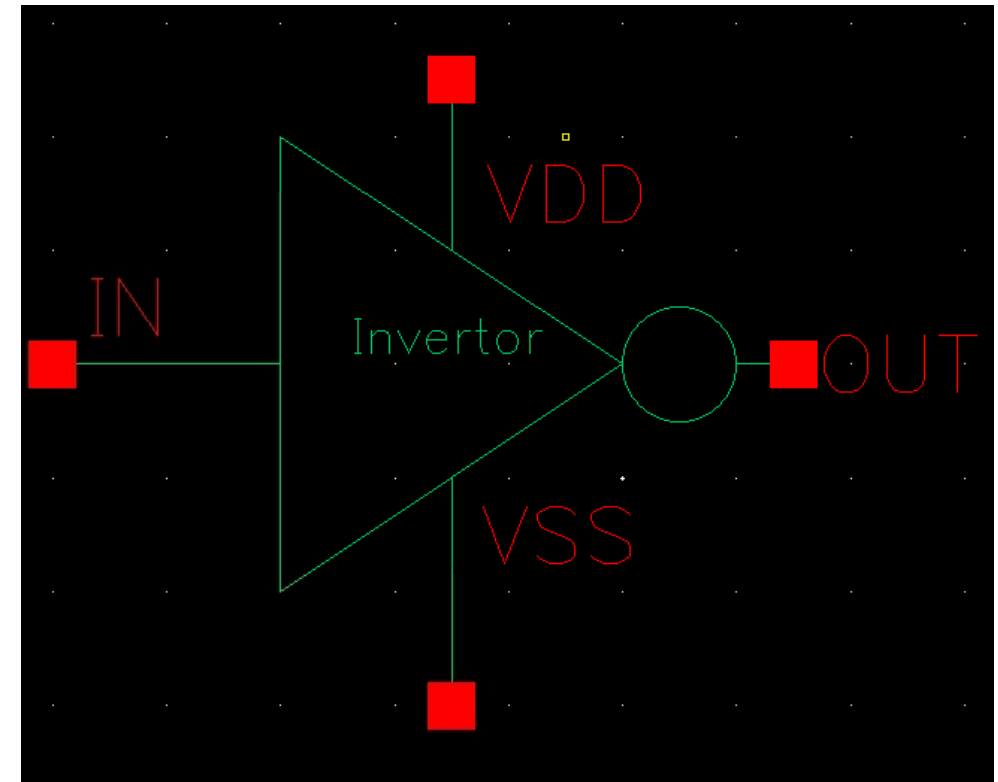
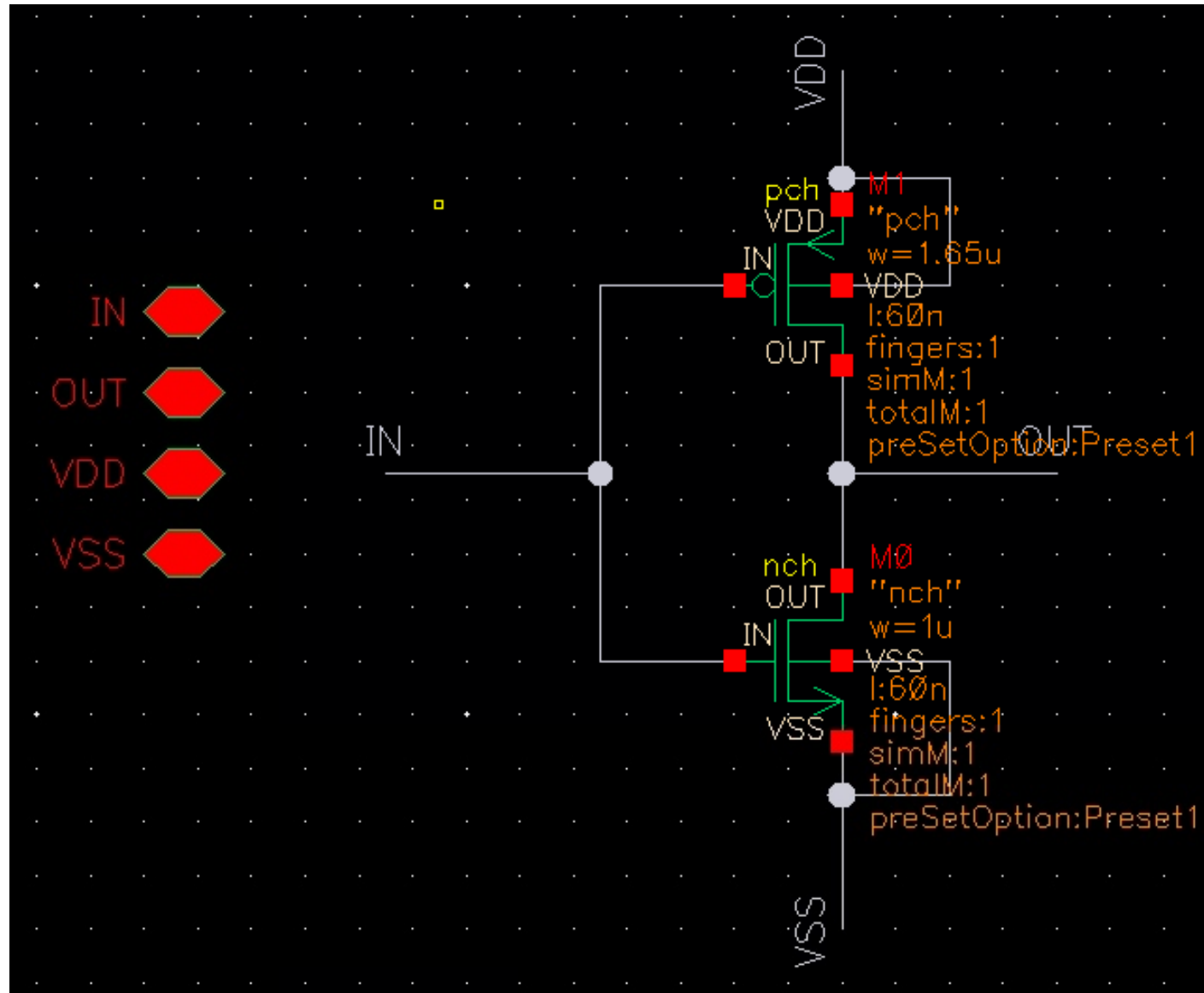
LAYOUT CELL NAME: Transmission_Gate
SOURCE CELL NAME: Transmission_Gate

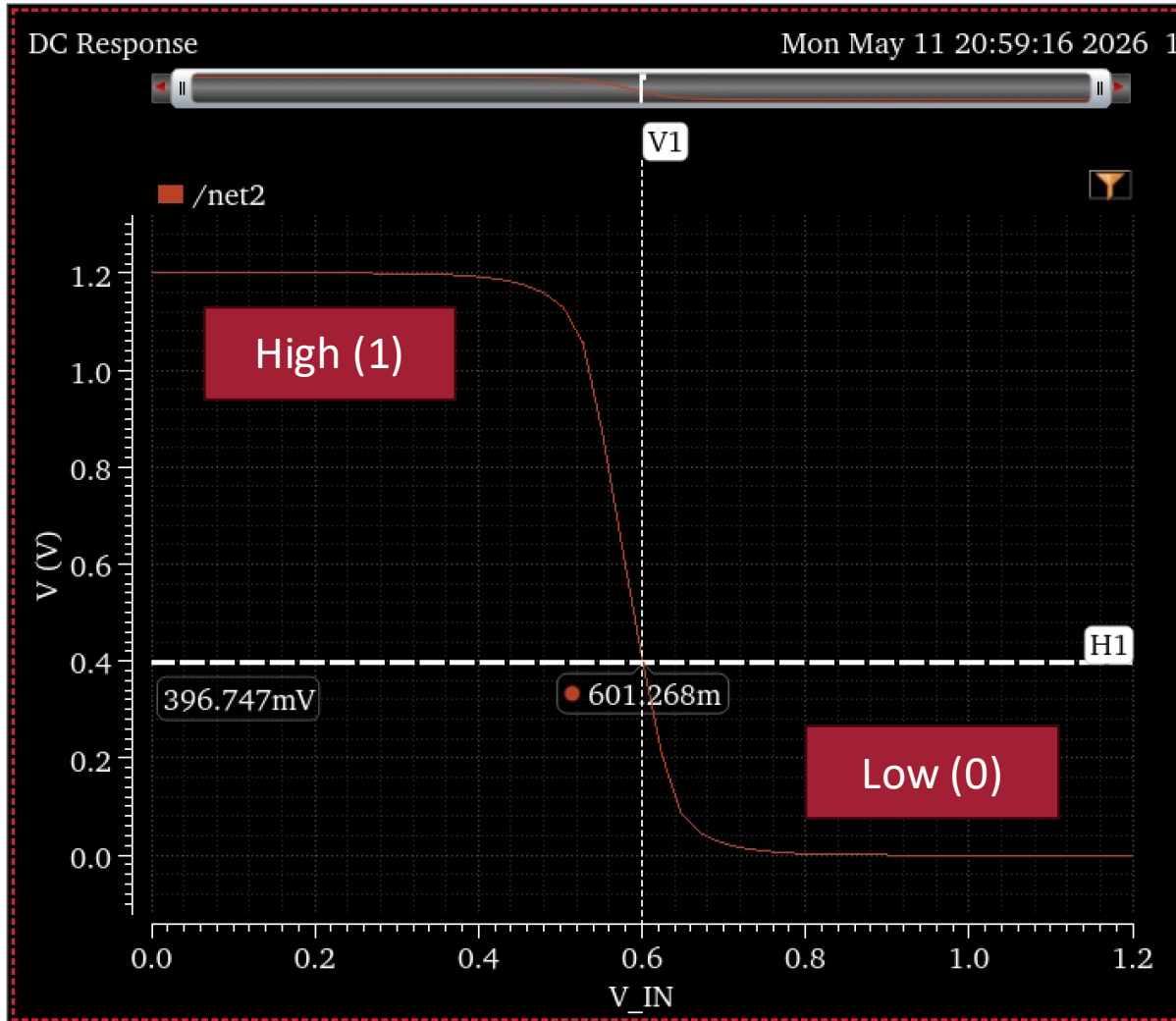
INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	5	5	
Nets:	6	6	
Instances:	2	2	MN (4 pins)
	2	2	MP (4 pins)
Total Inst:	4	4	

NUMBERS OF OBJECTS AFTER TRANSFORMATION

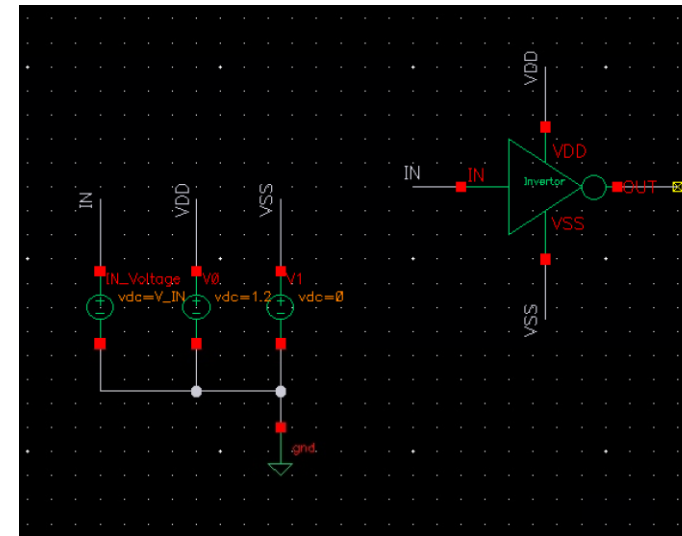
	Layout	Source	Component Type
--	--------	--------	----------------





Explanation:

The graph on the side indicates the V_{in} and V_{out} relationship of the inverter, which is used to show the symmetric nature between the highs and lows. The results show approximate symmetric behavior.



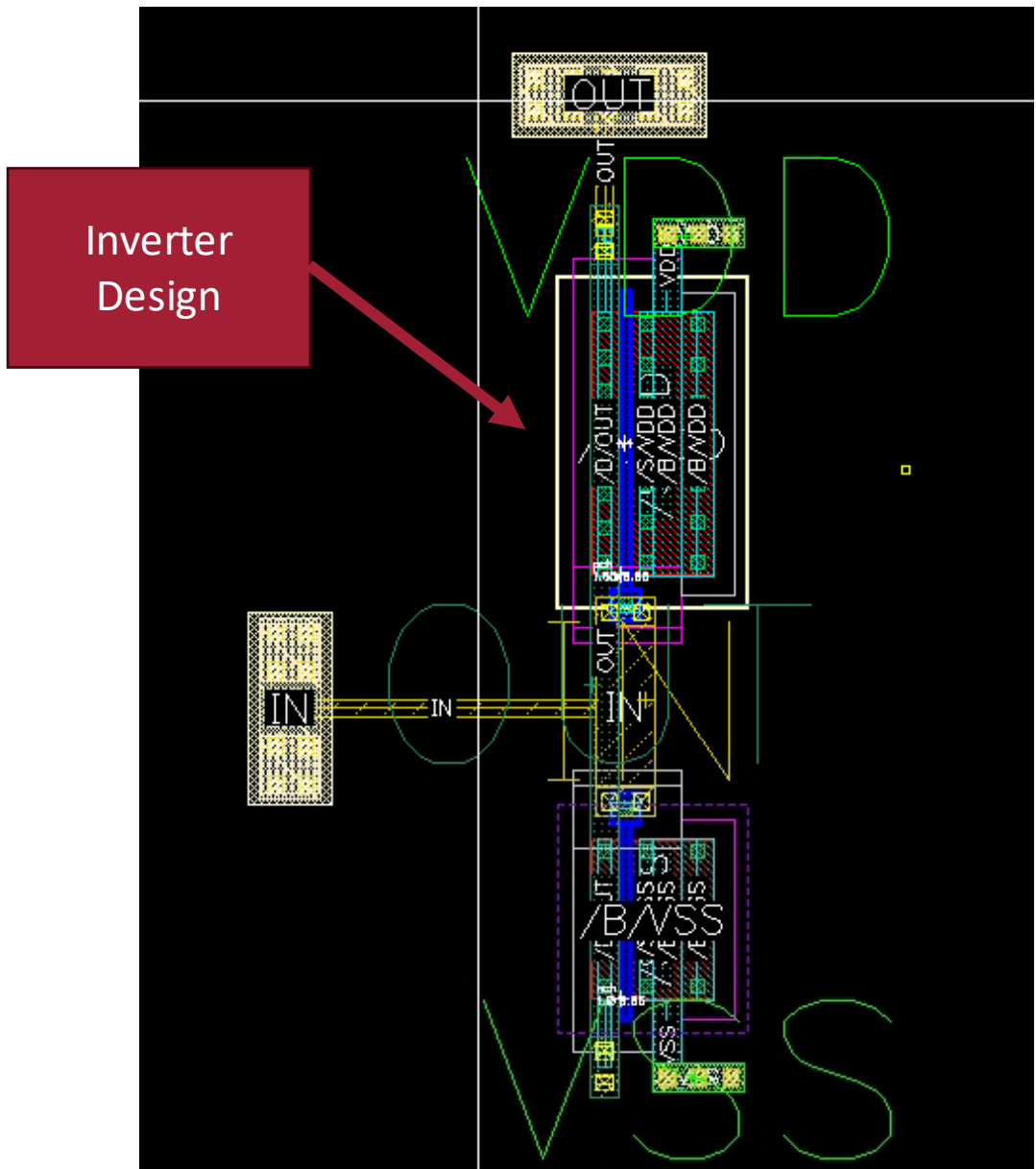
Analyses

☒ dc 0 1.2 Automatic Start-Stop

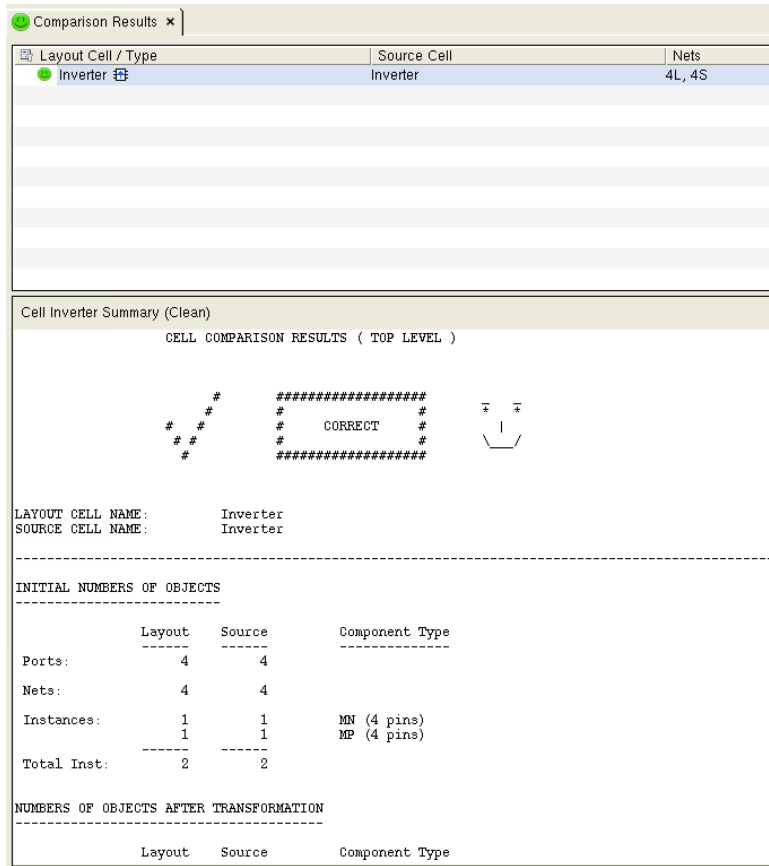
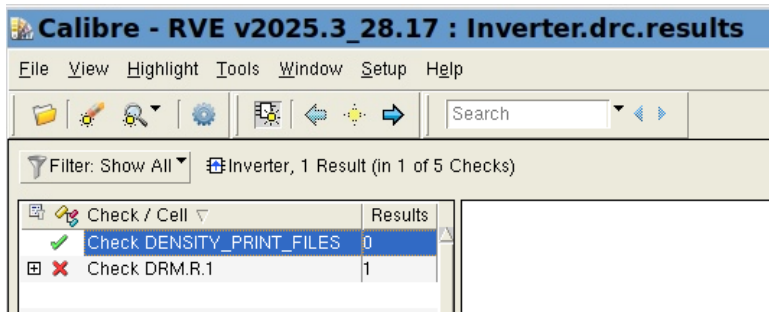
Click to add analysis

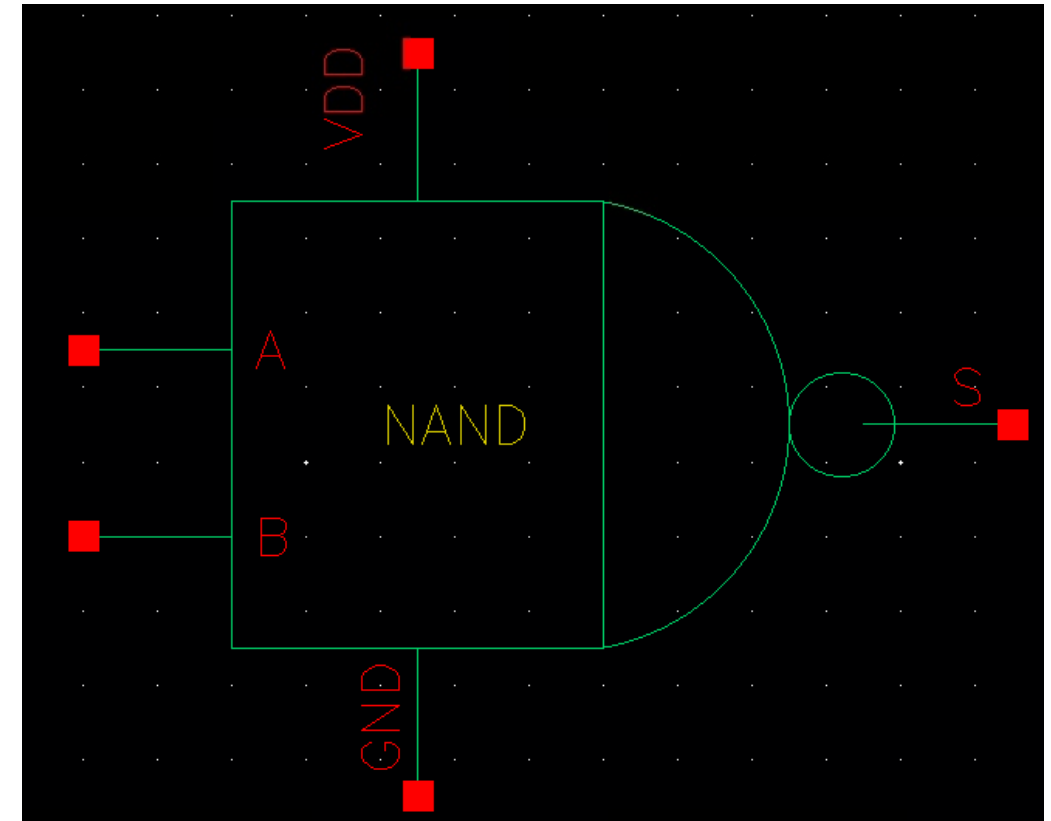
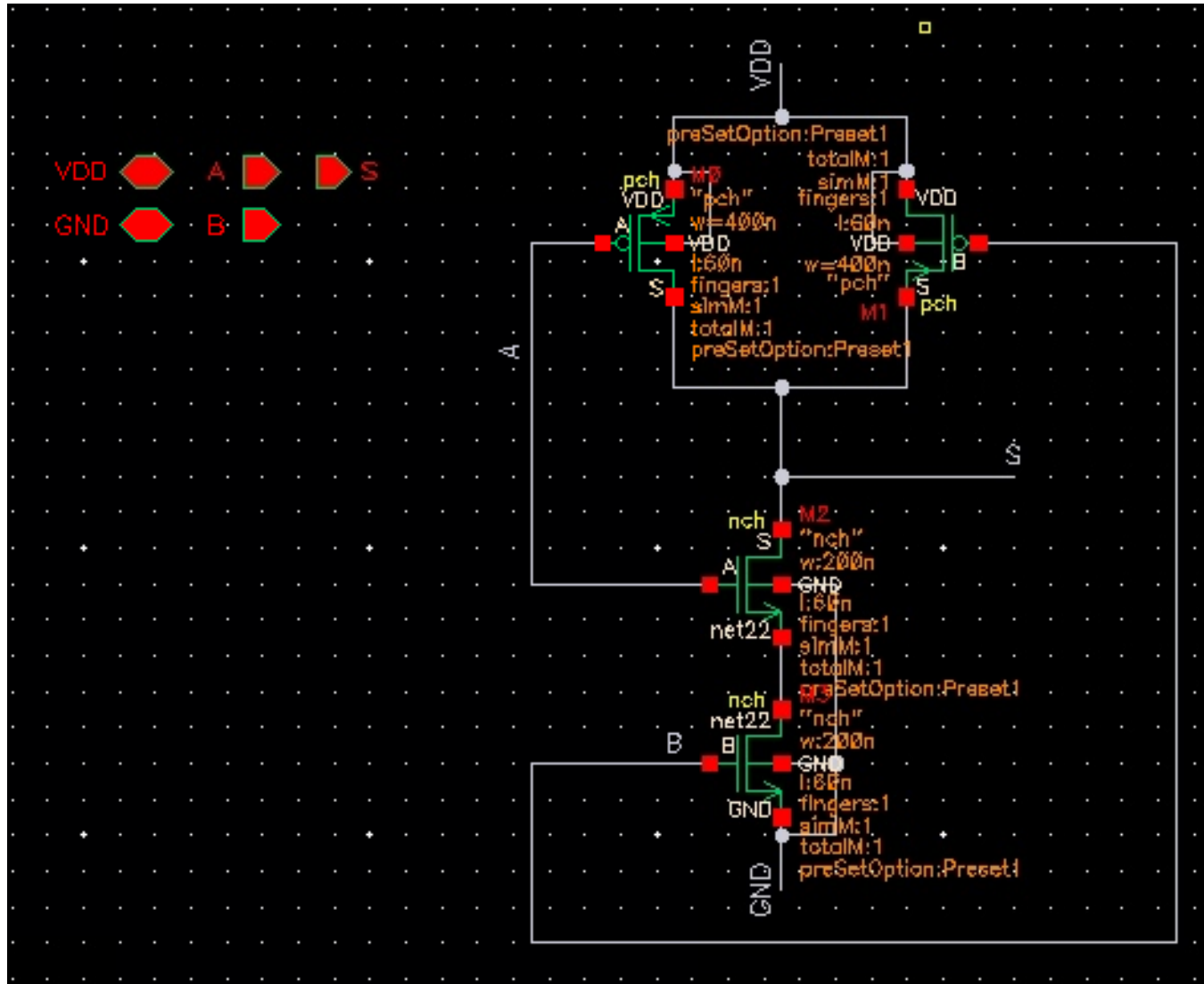
Design Variables

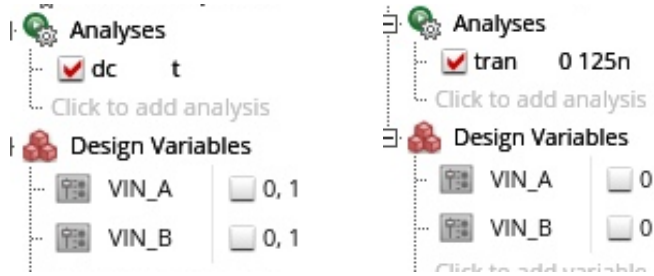
V_IN	0
WU	1.5u



Inverter - Image of Layout & DRC/LVS Clean







Proj_208_NAND_TB_2	prop_delay	expr	(cross(VT("/OUT3") 0.6 1 "rising" nil nil nil) - cross(VT("/OUT2") 0.6 1 "falling" nil nil nil))	point	☒	☐	< 10p
Proj_208_NAND_TB_2	rise_time	expr	(cross(VT("/OUT3") 1.08 1 "rising" nil nil nil) - cross(VT("/OUT3") 0.12 1 "rising" nil nil nil))	point	☒	☐	< 15p
Proj_208_NAND_TB_2	fall_time	expr	(cross(VT("/OUT3") 0.12 1 "falling" nil nil nil) - cross(VT("/OUT3") 1.08 1 "falling" nil nil nil))	point	☒	☐	< 15p
Parameters: VIN_A=0, VIN_B=0							
1	Proj_208_NAND_TB_1	VDC("/OUT")	1				
Parameters: VIN_A=0, VIN_B=1							
2	Proj_208_NAND_TB_1	VDC("/OUT")	1				
Parameters: VIN_A=1, VIN_B=0							
3	Proj_208_NAND_TB_1	VDC("/OUT")	1				
Parameters: VIN_A=1, VIN_B=1							
4	Proj_208_NAND_TB_1	VDC("/OUT")	490.7n				
Parameters: VIN_A=0, VIN_B=0							
5	Proj_208_NAND_TB_2	prop_delay	-50n	< 10p			pass
5	Proj_208_NAND_TB_2	rise_time	Retracted	< 15p			pass
5	Proj_208_NAND_TB_2	fall_time	Retracted	< 15p			pass

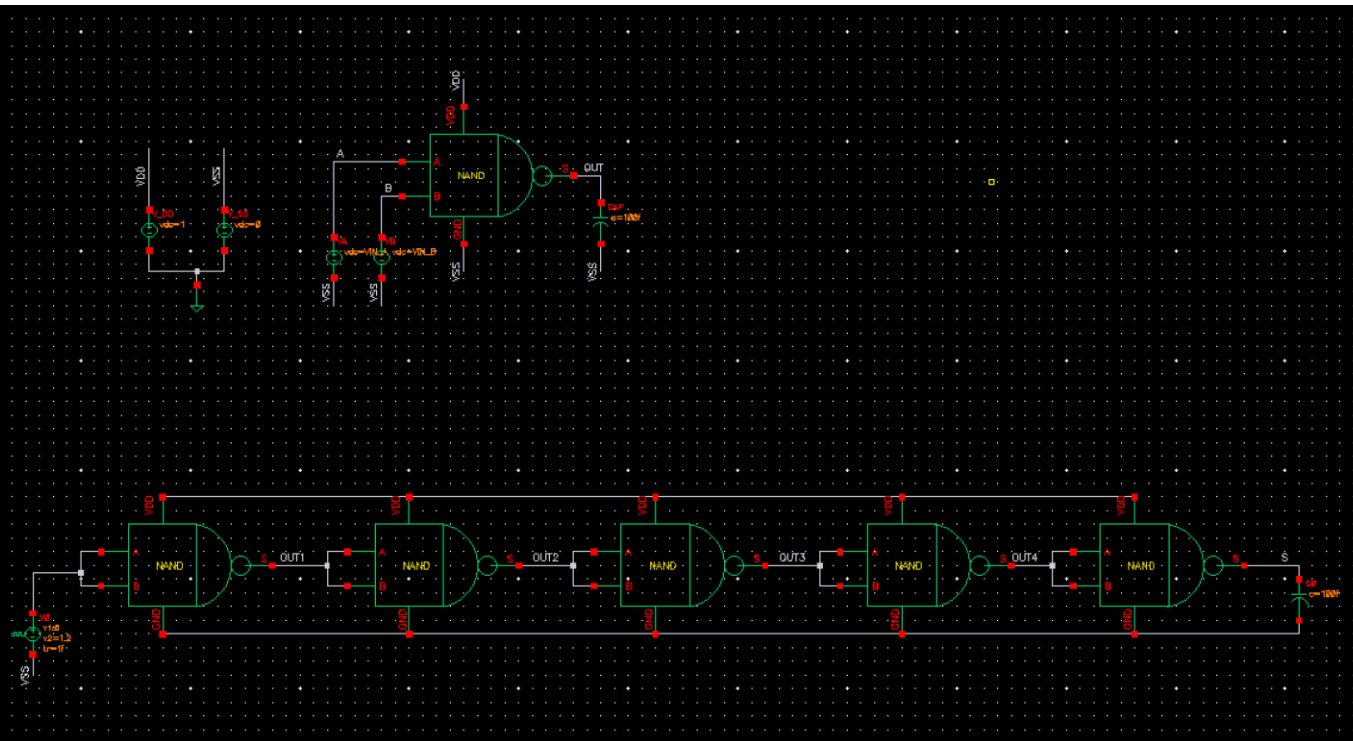
Explanation:

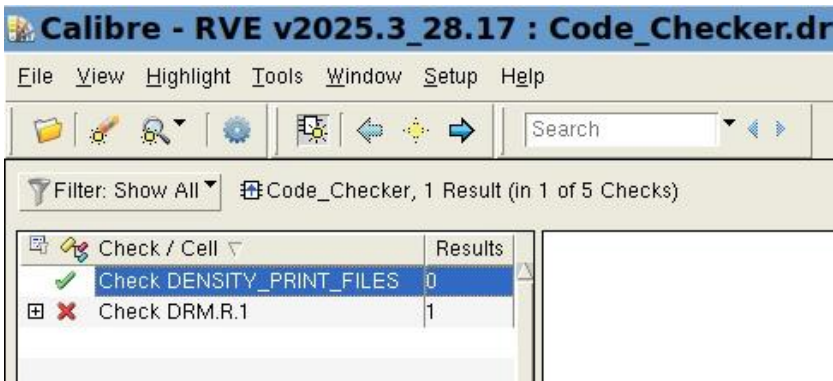
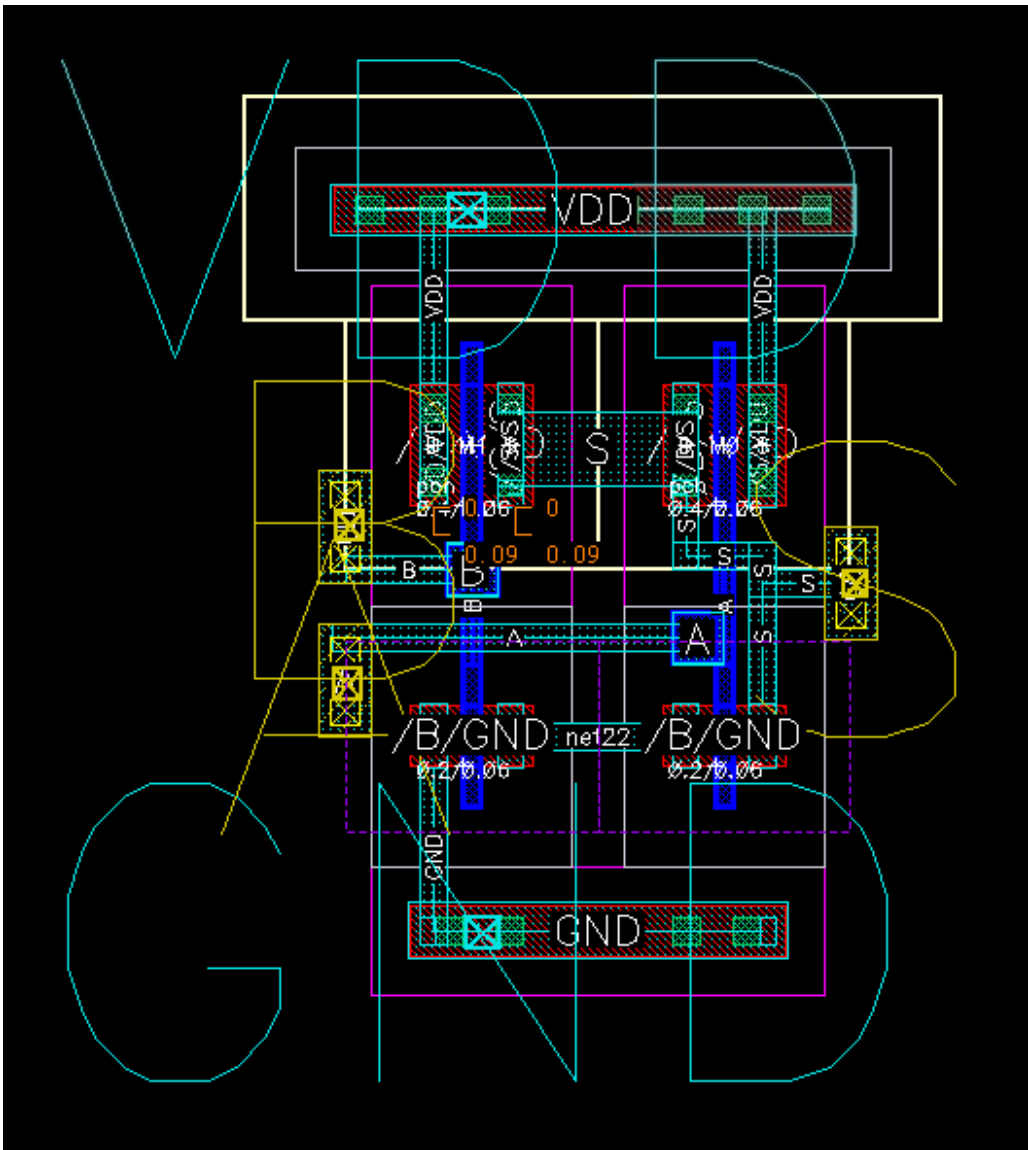
Logic - Test 1:

The NAND logic was tested, which showed that when both inputs are high, which causes the output to be relatively low, which is indicated by a nano-volts, and the other voltage were determined to be VDD

Time - Test 2:

The propagation delay (50n) was determined by the reaction time through the cascade of NAND gates. Furthermore, the rise time (800a) and fall time (678a) were determined, which all follow the requirements.





Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
Code_Checker	Code_Checker	14L, 14S	13L, 13S	7L, 7S

Cell Code_Checker Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

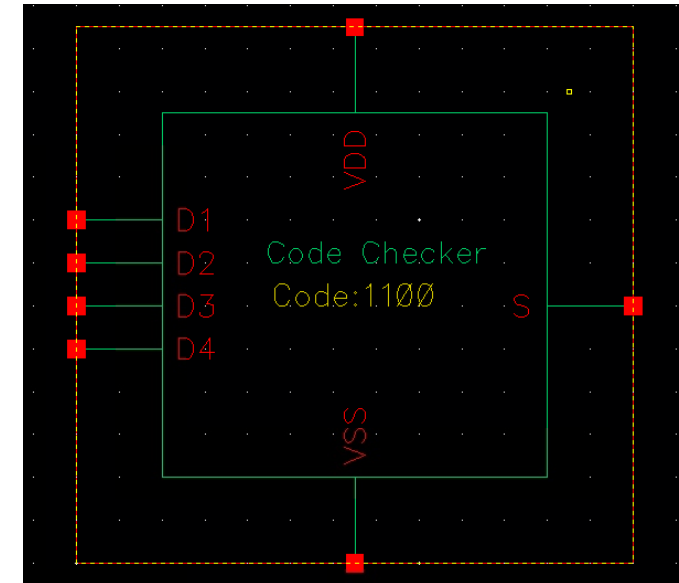
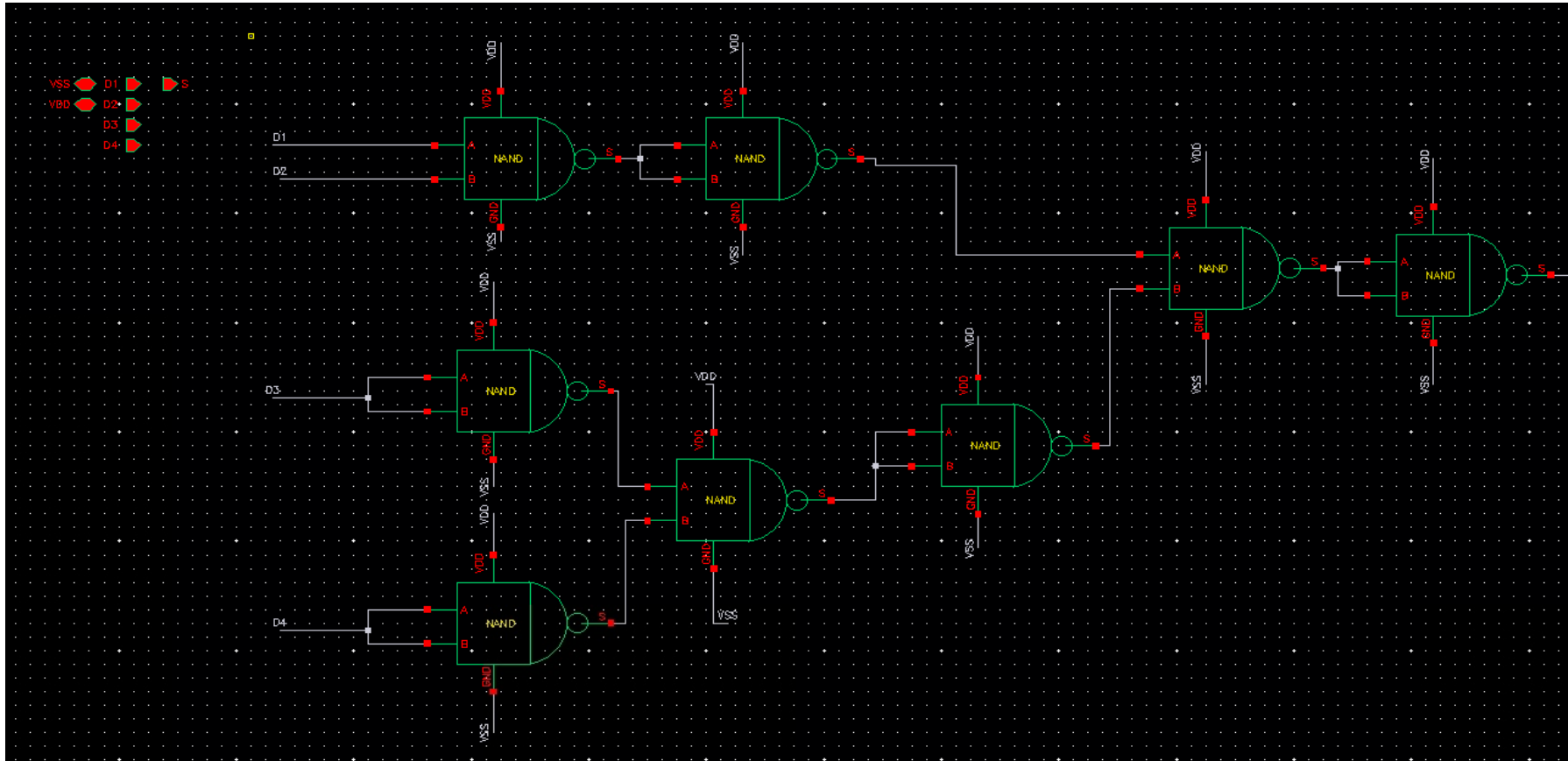
CORRECT #
#####

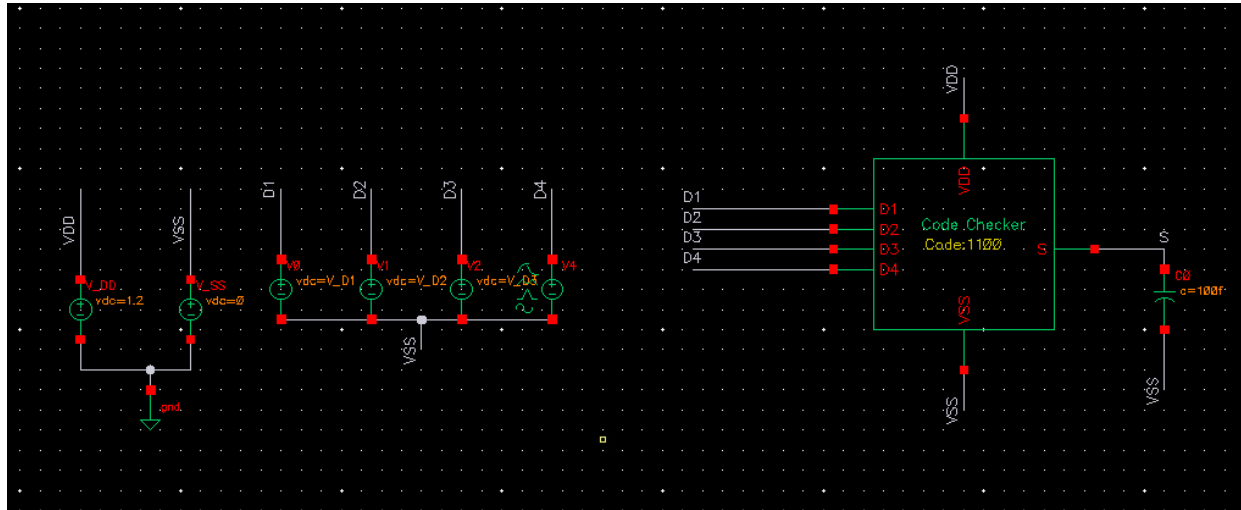
LAYOUT CELL NAME: Code_Checker
SOURCE CELL NAME: Code_Checker

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	7	7	
Nets:	22	22	
Instances:	16	16	MN (4 pins)
	16	16	MP (4 pins)
Total Inst:	32	32	

NUMBERS OF OBJECTS AFTER TRANSFORMATION





Explanation:

The propagation delay was determined through the perturbation of the input and then determine the delay in response in the output to determine the delay. The delay was determined to be 347.9 p second, which is within the required range of less than 500ps

Name	Type	Details	Value	Plot	Save	Spec
	signal	/S		☒	☒	
	signal	/D4		☒	☒	
	expr	(cross(VT("/S") 0.6 1 "rising" nil n...	347.9p	☒	☐	< 500p

Analyses

☒ tran 0.0125n

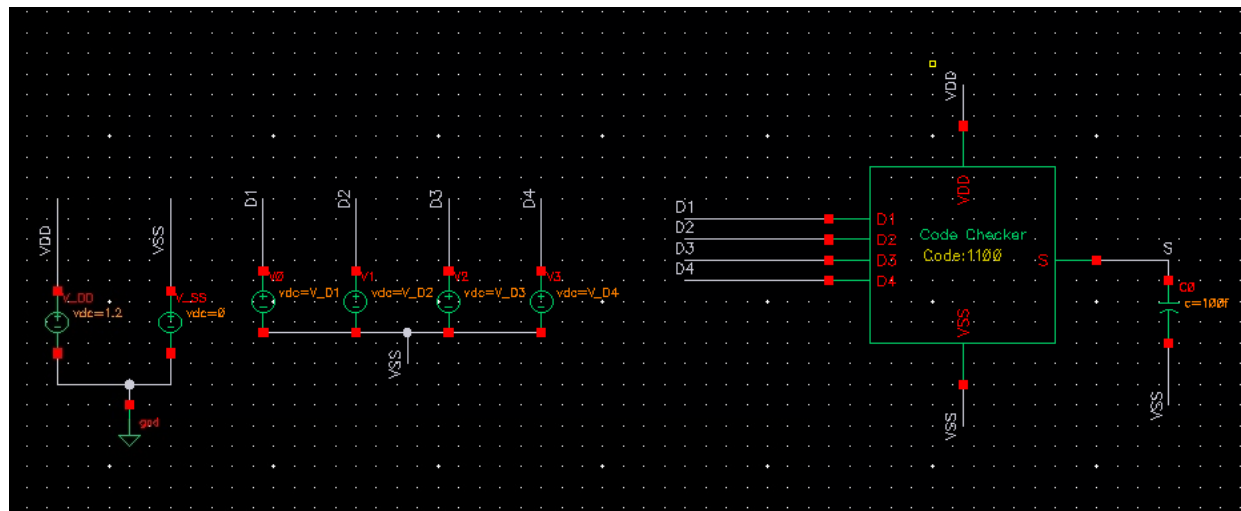
Click to add analysis

Design Variables

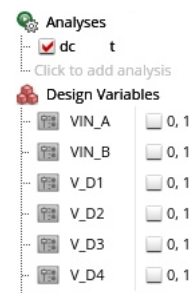
☐ V_D1 1

☐ V_D2 1

☐ V_D3 0

**Explanation:**

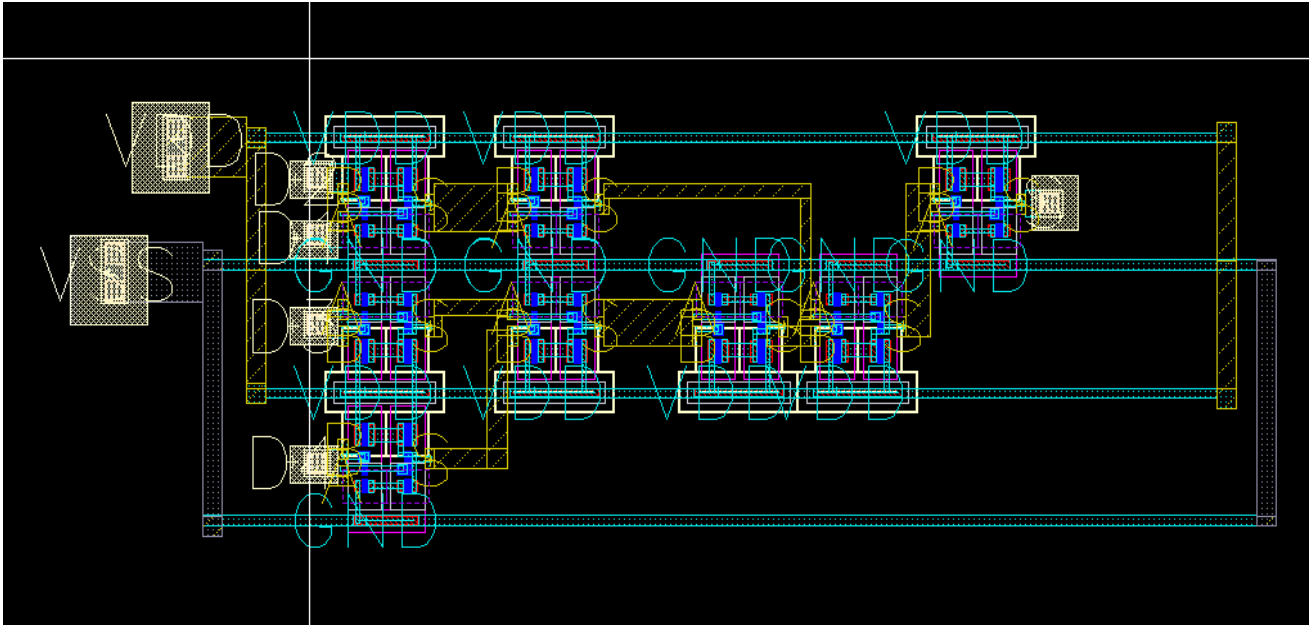
The desired code 1100 is required to send a signal of one, and the other codes are required to send a low signal, which is approximately 545.9nV.



Parameters: V_D1=0, V_D2=0, V_D3=0, V_D4=0		
1	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=0, V_D2=0, V_D3=0, V_D4=1		
2	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=0, V_D2=0, V_D3=1, V_D4=0		
3	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=0, V_D2=0, V_D3=1, V_D4=1		
4	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=0, V_D2=1, V_D3=0, V_D4=0		
5	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=0, V_D2=1, V_D3=0, V_D4=1		
6	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=0, V_D2=1, V_D3=1, V_D4=0		
7	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=0, V_D2=1, V_D3=1, V_D4=1		
8	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=1, V_D2=0, V_D3=0, V_D4=0		
9	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=1, V_D2=0, V_D3=0, V_D4=1		
10	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=1, V_D2=0, V_D3=1, V_D4=0		
11	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=1, V_D2=0, V_D3=1, V_D4=1		
12	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=1, V_D2=1, V_D3=0, V_D4=0		
13	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=1, V_D2=1, V_D3=0, V_D4=1		
14	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=1, V_D2=1, V_D3=1, V_D4=0		
15	Proj_208_Code_Check...	VDC("S")
Parameters: V_D1=1, V_D2=1, V_D3=1, V_D4=1		
16	Proj_208_Code_Check...	VDC("S")



Image of layout for full code checker & DRC/LVS Clean



Calibre - RVE v2025.3_28.17 : Code_Checker.dr

File View Highlight Tools Window Setup Help

Filter: Show All Code_Checker, 1 Result (in 1 of 5 Checks)

Check / Cell	Results
Check DENSITY_PRINT_FILES	0
Check DRM.R.1	1

Comparison Results x

Layout Cell / Type	Source Cell	Nets	Instances	Ports
Code_Checker	Code_Checker	14L, 14S	13L, 13S	7L, 7S

Cell Code_Checker Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

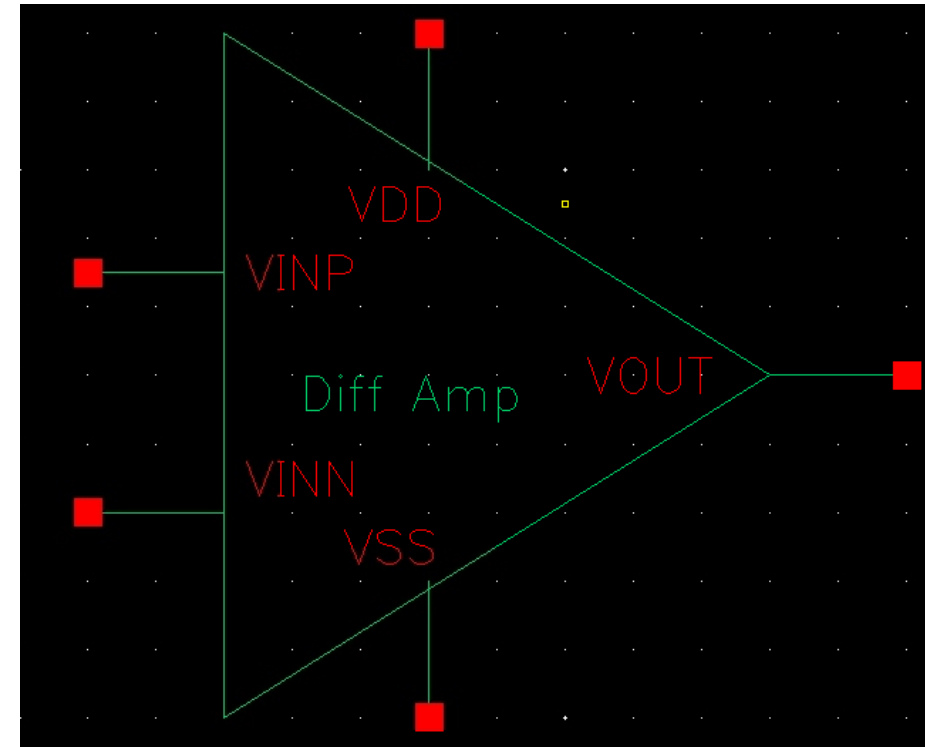
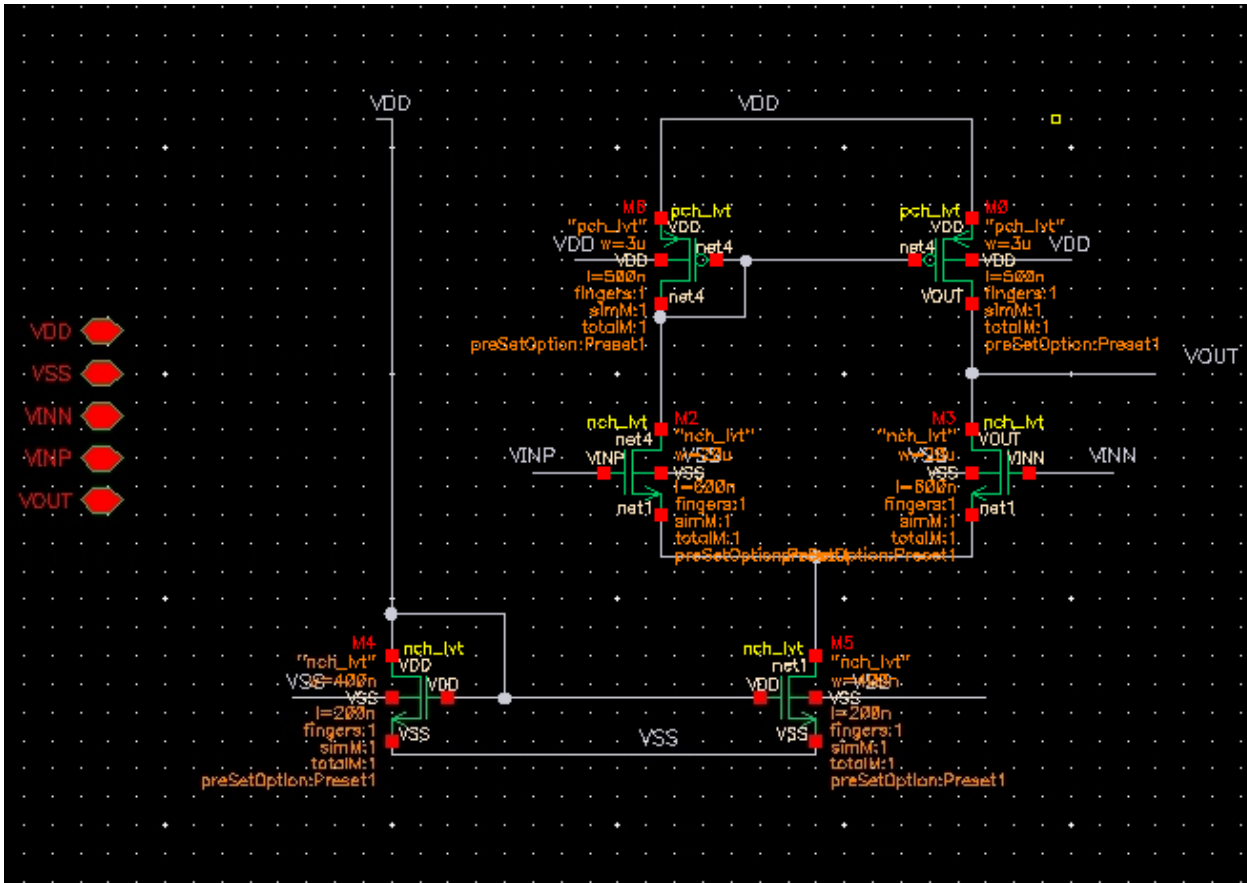
CORRECT #
#####

LAYOUT CELL NAME: Code_Checker
SOURCE CELL NAME: Code_Checker

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	7	7	
Nets:	22	22	
Instances:	16	16	MN (4 pins)
	16	16	MP (4 pins)
Total Inst:	32	32	

NUMBERS OF OBJECTS AFTER TRANSFORMATION



Explanation:

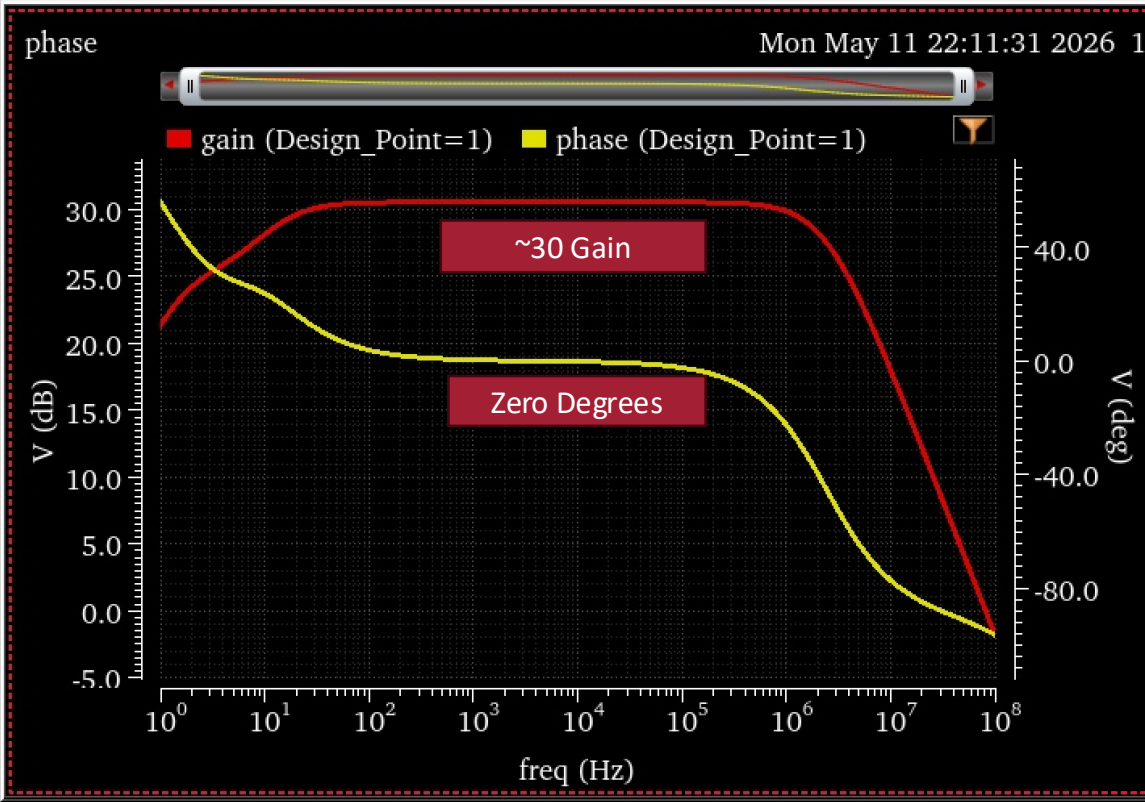
Single-ended output differential amplifier topology was deployed, while using a current mirror to make the amplification maximal.



Schematic AC

ac_tb	gain	expr	db(vfreq('ac "/VOUT"))	point	☒	☐	
ac_tb	phase	expr	phaseDegUnwrapped(vfreq('ac ...	point	☒	☐	
ac_tb	gain_value	expr	value(gain 1000000.0)	point	☒	☐	> 26
ac_tb	power_consumption	expr	(IDC("/V3/PLUS") * VDC("/VDD"))	point	☒	☐	< 500u

ac_tb	gain				
ac_tb	phase				
ac_tb	gain_value	29.89	> 26	pass	
ac_tb	power_consumption	-261.5u	< 500u	pass	



Explanation:

The gain of the system was determined through the selection of gain at 1Mhz frequency. Which was determined to be approximately 29.89 dB, and the power consumption was determined to be on the magnitude of 261.5 uW.

The image on the left indicates the frequency range and the corresponding gain and phase at each frequency.



Schematic Swing

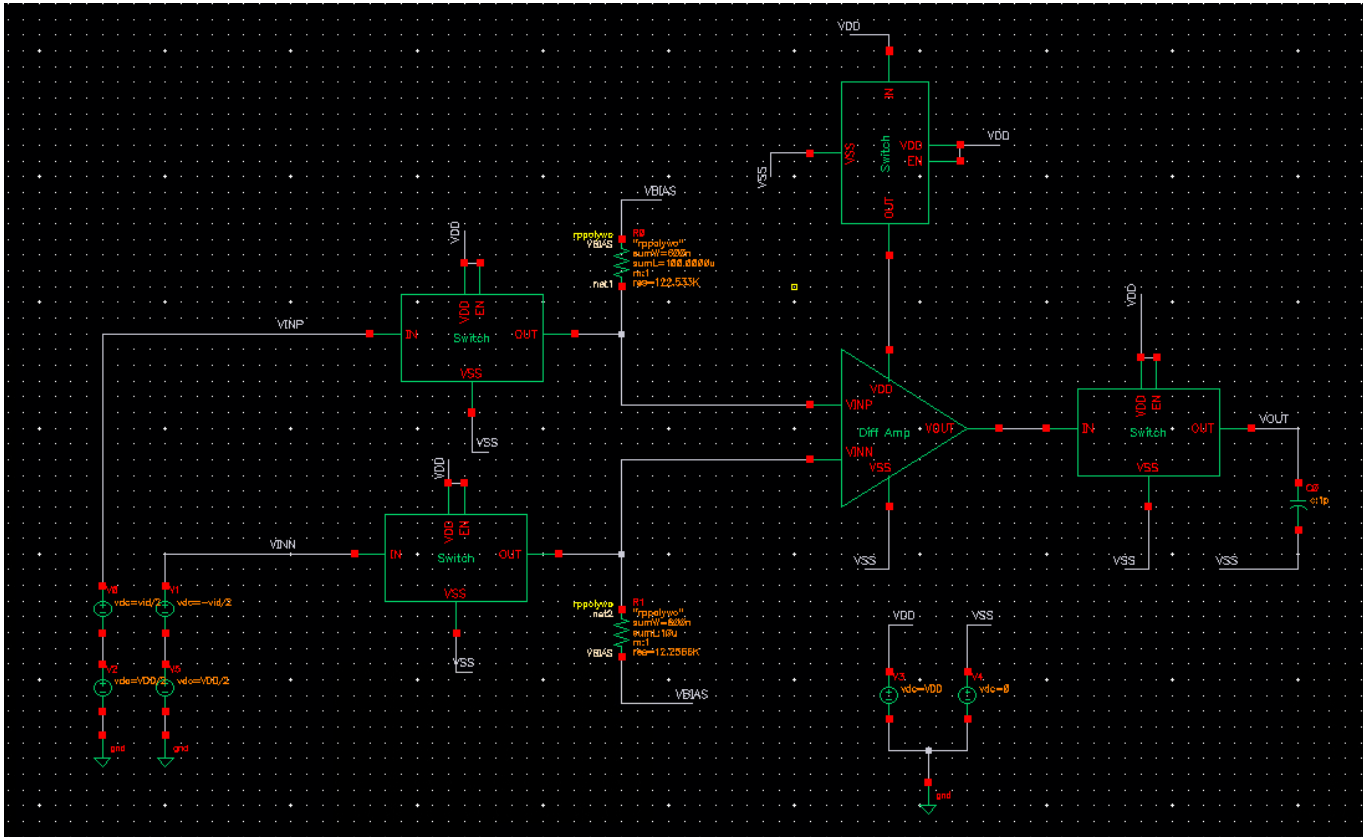
Analyses

☒ dc t -15m 15m 0.1m Linear Step Size Start-Stop

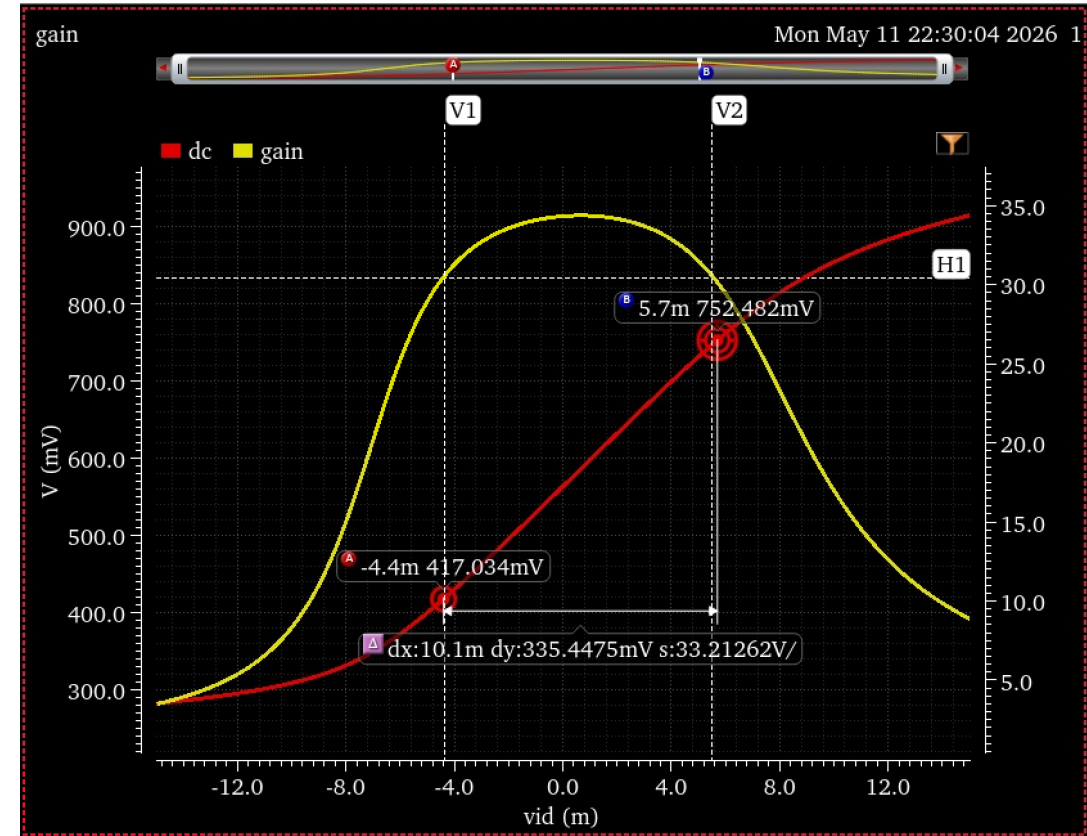
Click to add analysis

Design Variables

VDD	1.2
vid	1



Simulation setup of Swing



Explanation:

The above shows the process of determining the swing of the device, which is the interval of voltage change between 10% loss of amplification, which was determined to be dy , which has the value of 335.33 mV



Schematic Transient

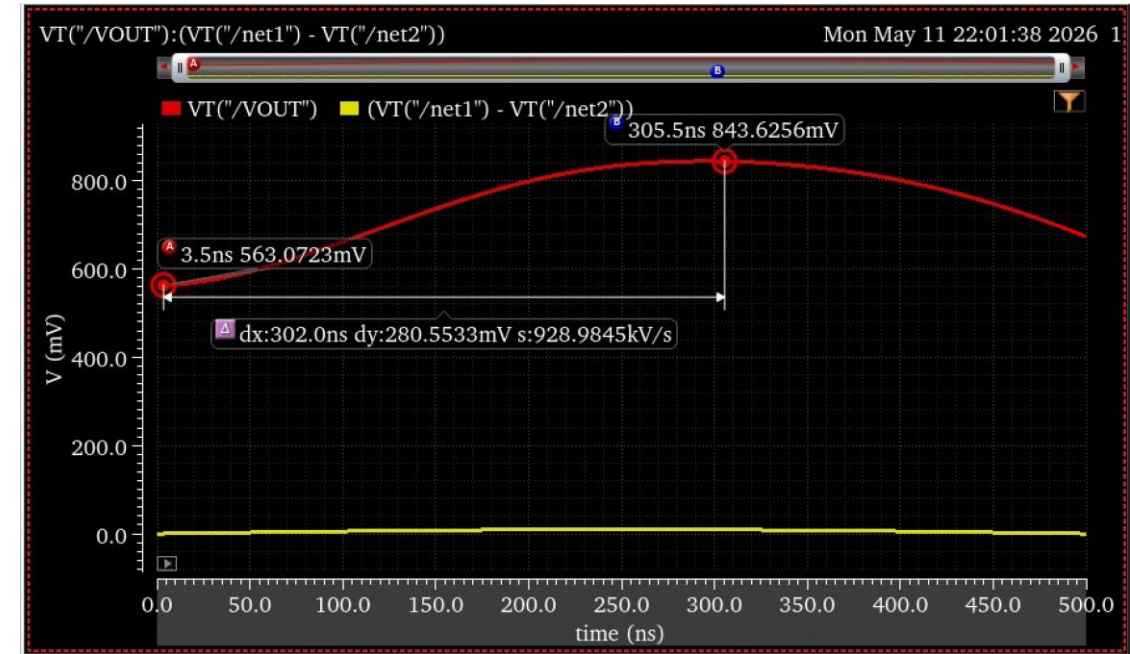
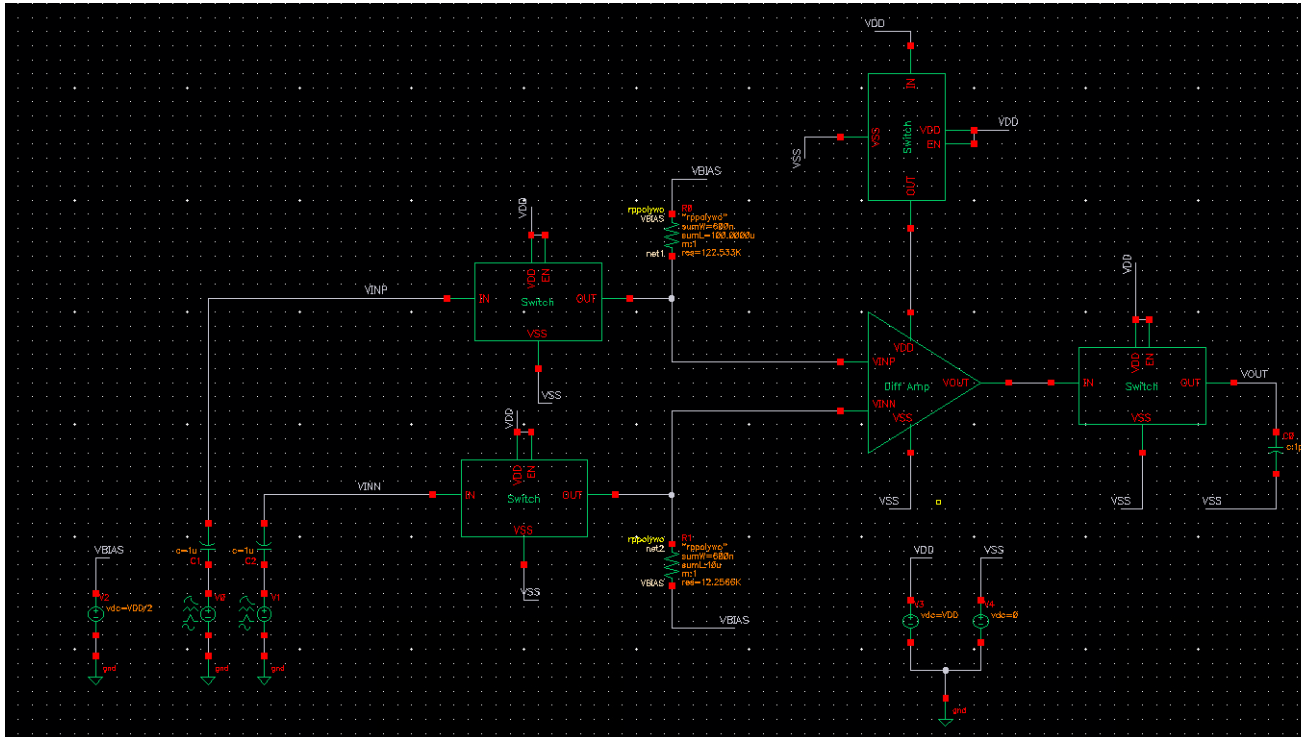
Analyses

☒ tran 0 0.5u moderate

Click to add analysis

Design Variables

fs	1M
VDD	1.2
vs	5m



Explanation:

The above shows the transient response of the amplifier, which shows that the signal is indeed amplified as shown by the higher amplitude in the response compared to the difference input.



Summary of Results:

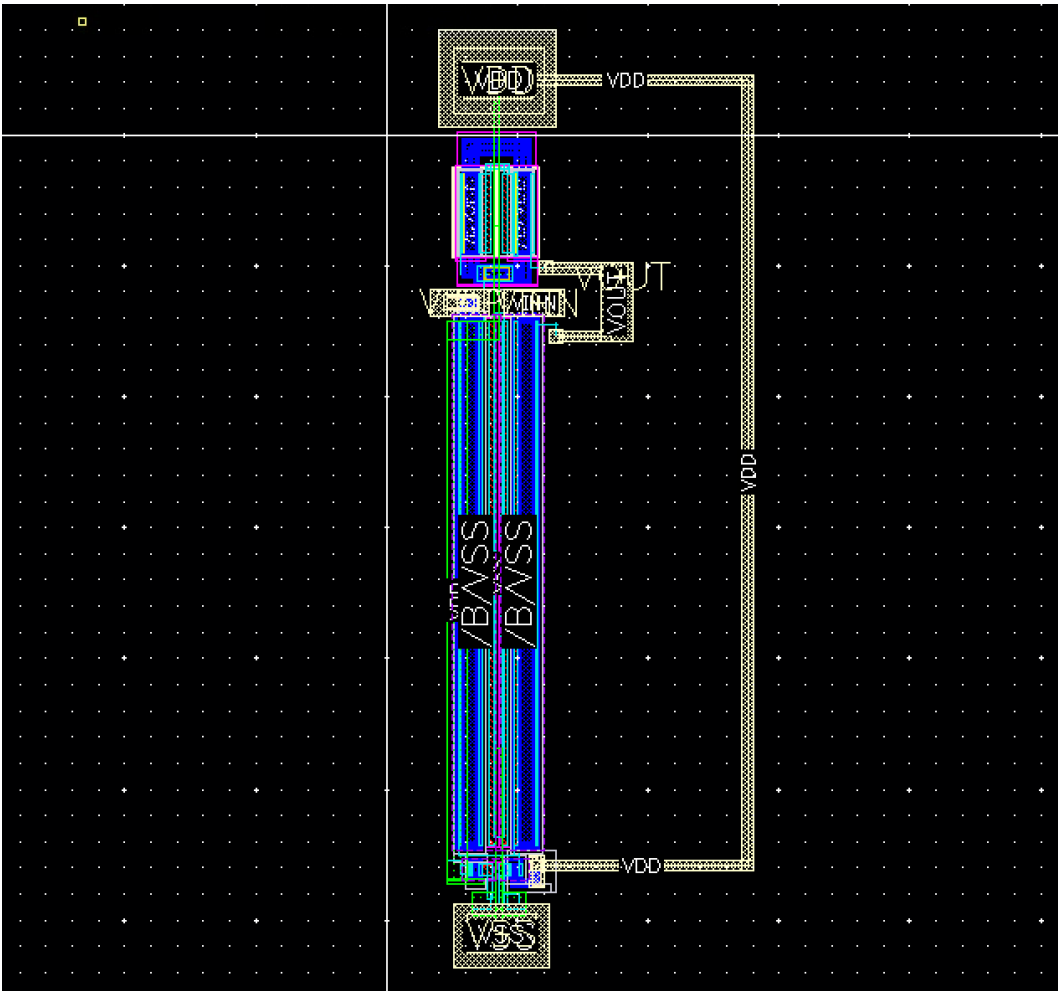
ac_tb	gain				
ac_tb	phase				
ac_tb	gain_value	29.89	> 26		pass
ac_tb	power_consumption	-261.5u	< 500u		pass

Specification		Your Design
Gain (@ 1MHz)	>26 dB	28.89 dB
Power Consumption	<500 uW	261.5uW
Peak-to-peak output swing	>300 mV	335 mV

Explanation:

The above displays the concise depiction of the results. The gain, power consumption, and the peak-to-peak output swing in voltage.





Calibre - RVE v2025.3_28.17 : DIFF_Amp.drc.results

File View Highlight Tools Window Setup Help

Filter: Show All | DIFF_Amp, 1 Result (in 1 of 5 Checks)

Check / Cell	Results
Check DENSITY_PRINT_FILES	0
Check DRM.R.1	1

Comparison Results

Layout Cell / Type	Source Cell	Nets
DIFF_Amp	DIFF_Amp	7L, 7S

Cell DIFF_Amp Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

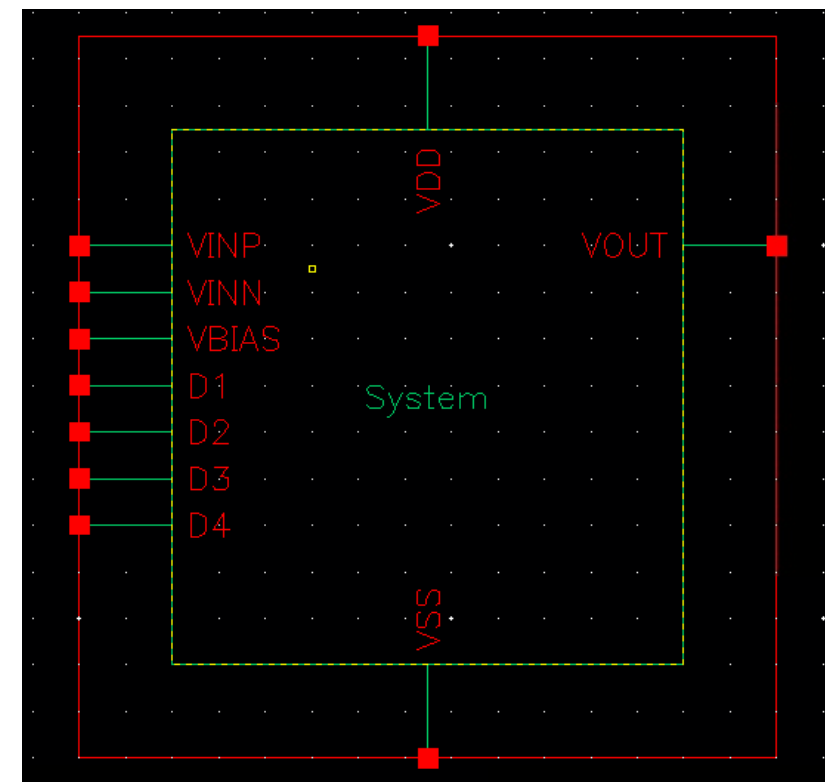
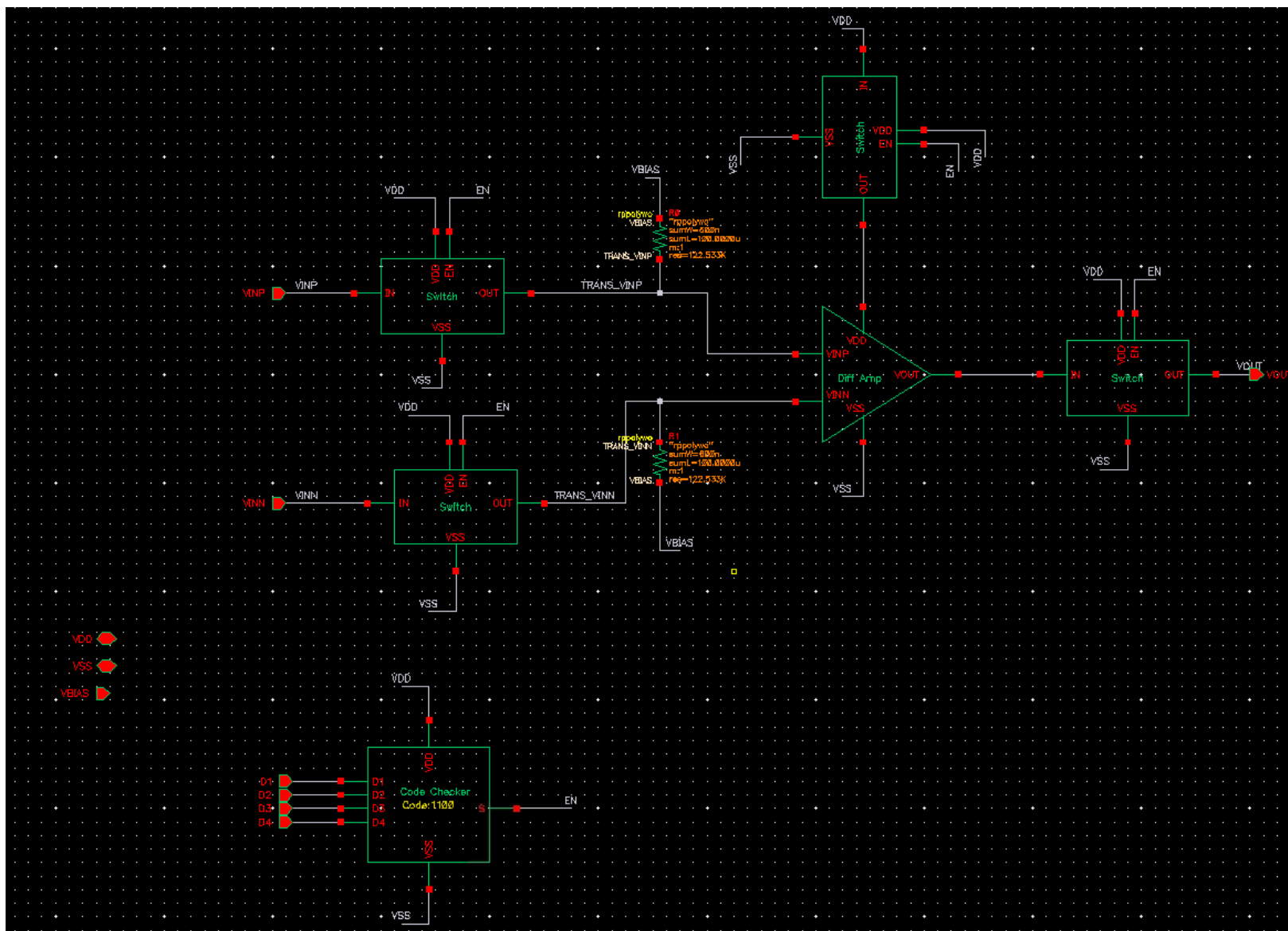
```
#####
# CORRECT #
#####
```

LAYOUT CELL NAME: DIFF_Amp
SOURCE CELL NAME: DIFF_Amp

NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	5	5	
Nets:	7	7	
Instances:	4	4	MN (4 pins)
	2	2	MP (4 pins)
Total Inst:	6	6	

***** INFORMATION AND WARNINGS *****





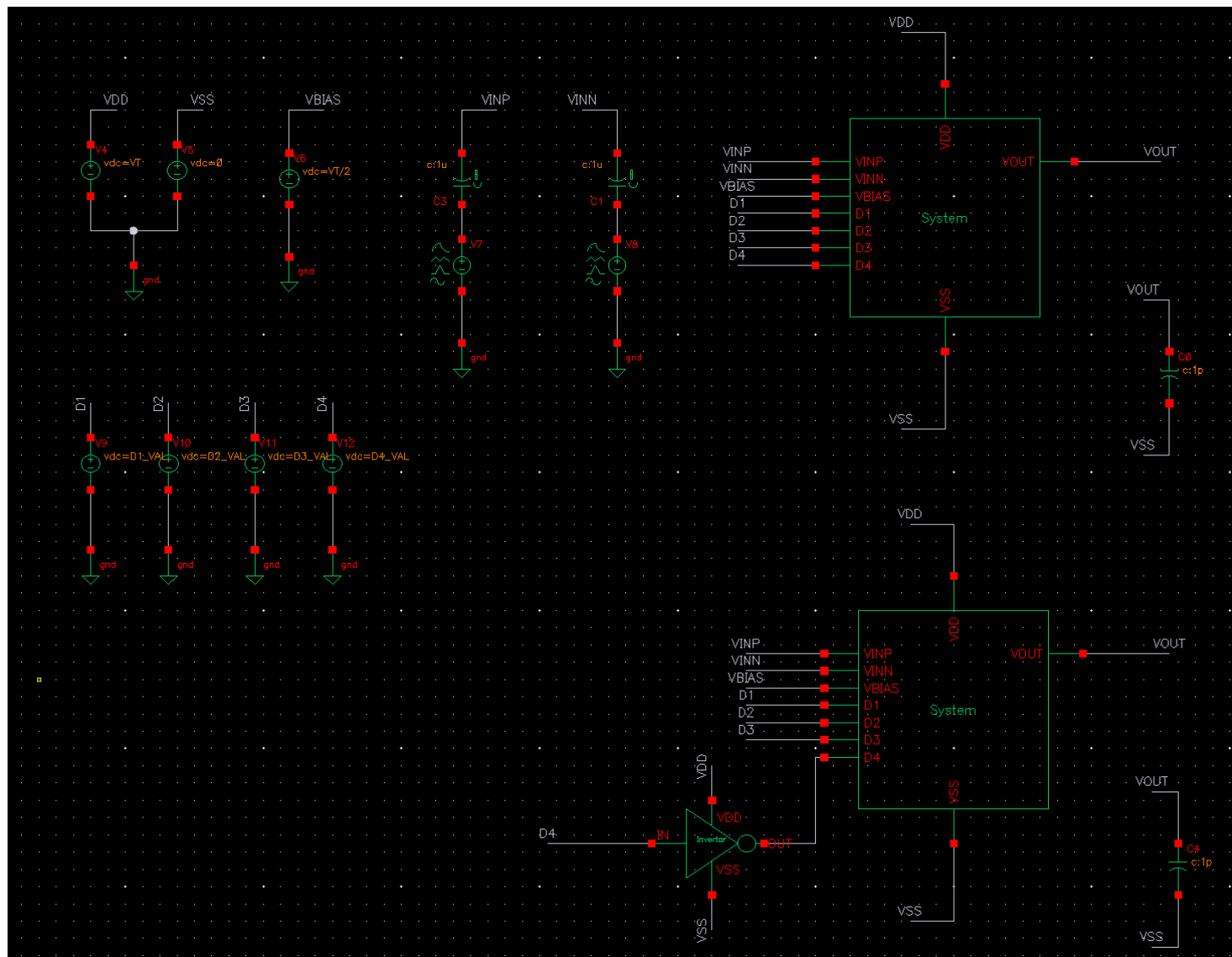
Analyses

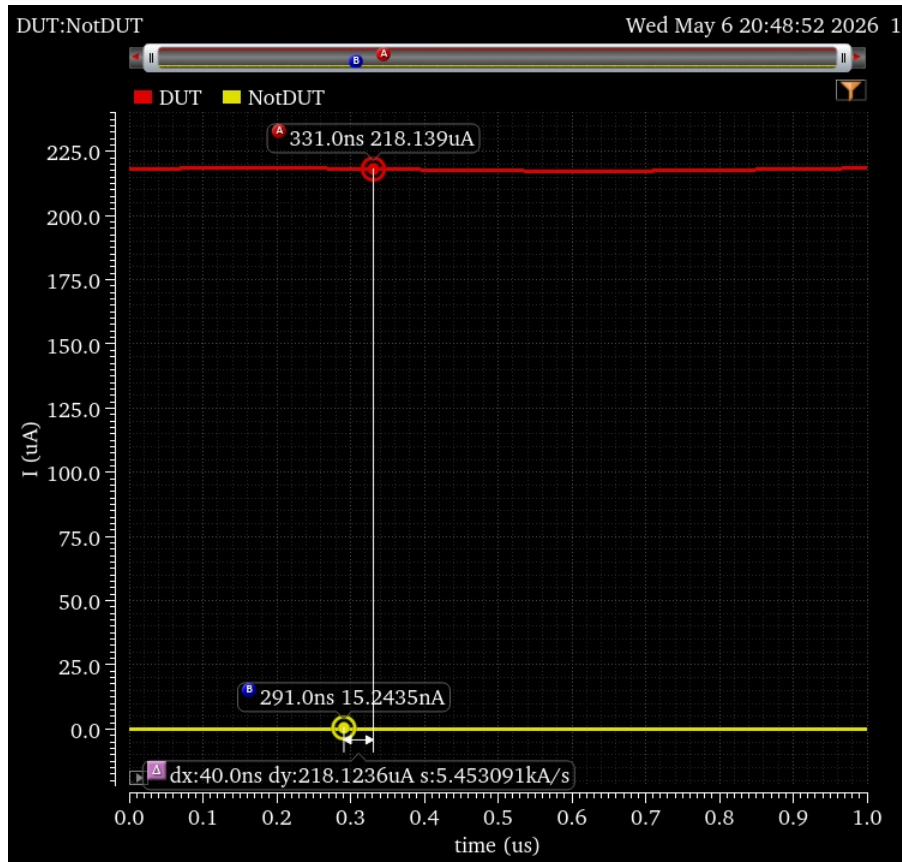
- ☒ tran 0 1u moderate
- ☐ ac 1 100M 20 Logarithmic Points Per Decade St..

Click to add analysis

Design Variables

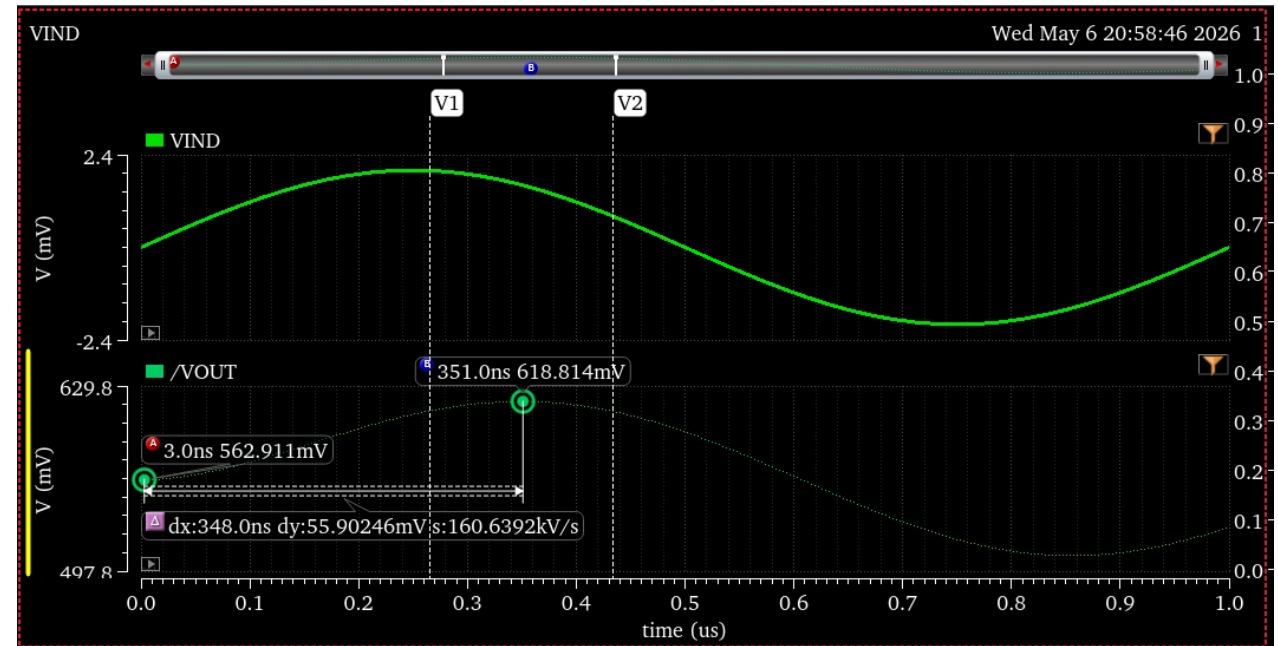
D1_VAL	1.2
D2_VAL	1.2
D3_VAL	0
D4_VAL	0
f1	1M
v1	1m
VT	1.2





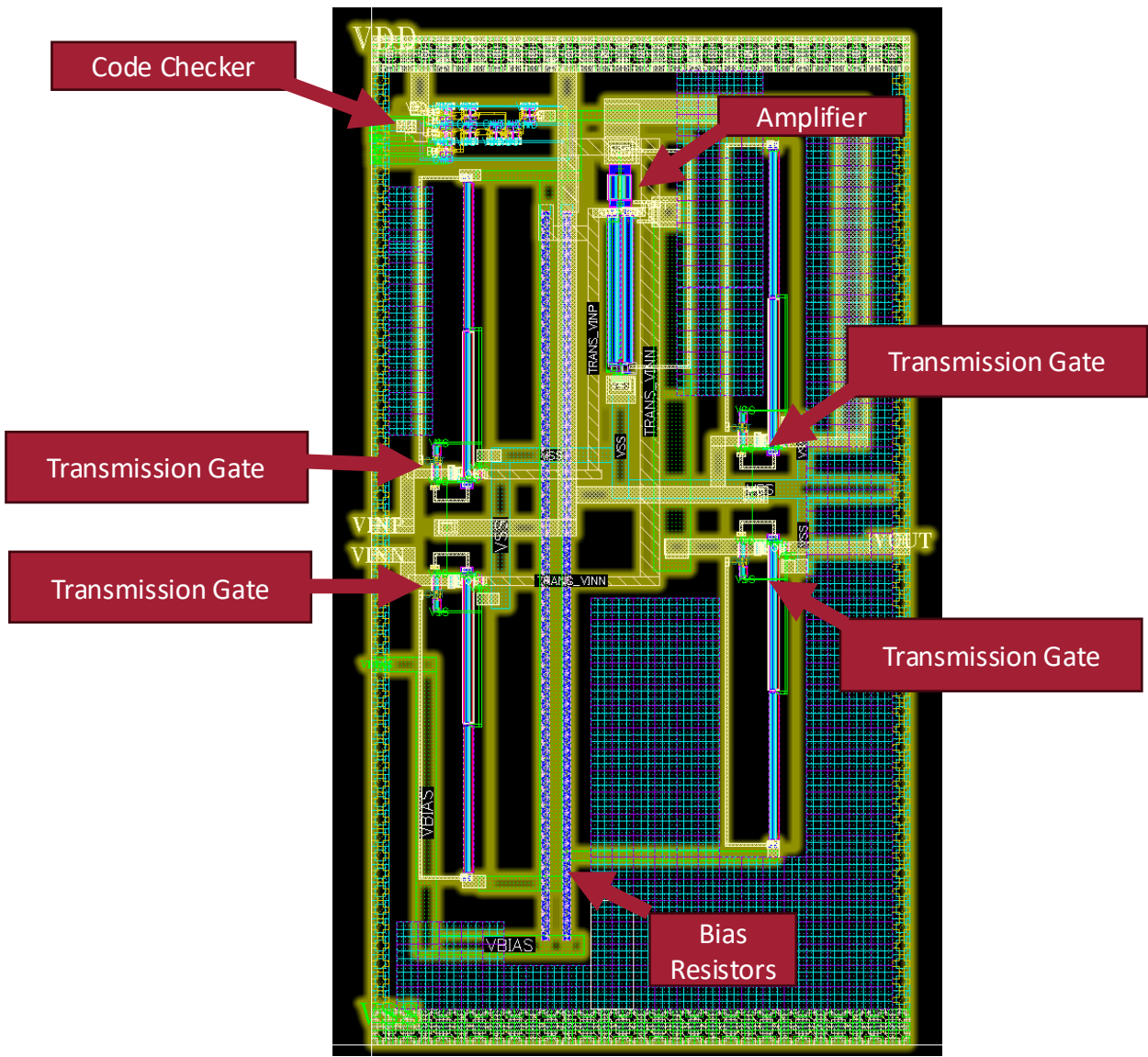
Explanation:

The above displays the quiescent current when turned on (~218.139 uA) and the quiescent current when turned off (~15.243 nA)



Explanation:

The above shows the transient response of the full system, which given that the input is 2mA, then the gain of the system is about $20 \times \log_{10}(27.5) \approx 28.79$ dB, which agrees with the Spec



Calibre - RVE v2025.3_28.17 : Full_Sys.drc.results

File View Highlight Tools Window Setup Help

Filter: Show All Full_Sys, 11 Results (in 10 of 14 Checks)

Check / Cell	Results
Check DENSITY_PRINT_FILES	0
Check DRM.R.1	1
Check M2.DN.1	1
Check M3.DN.1	1
Check M4.DN.1	1
Check M5.DN.1	1
Check M6.DN.1	1
Check M8.DN.1	1
Check M9.DN.2	1
Check OD.DN.2	1
Check PO.DN.2	2

```
#####
#
#   CALIBRE SYSTEM
#
#   LVS REPORT
#
#####

REPORT FILE NAME: Full_Sys.lvs.report
LAYOUT NAME: Full_Sys.sp ('Full_Sys')
SOURCE NAME: Full_Sys.src.net ('Full_Sys')
RULE FILE: _calibre.lvs
CREATION TIME: Wed May 6 19:29:02 2026
CURRENT DIRECTORY: /export/home2/ocerv6/TSMC65/Calibre_Run
USER NAME: ocerv6
CALIBRE VERSION: v2025.3_28.17 Fri Aug 1 15:10:27 PDT 2025

OVERALL COMPARISON RESULTS

#####
#
#   CORRECT
#
#####

-----
Result      Layout      Source
-----
CORRECT     Full_Sys    Full_Sys
-----
```